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Adaptive distributed observer-based cooperative control for noncanonical nonlinear multi-agent systems modeled by RHONNs

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ABSTRACT

Different from traditional distributed control for multi-agent systems (MASs) composed of linear systems or nonlinear systems in canonical form, for MASs where followers have noncanonical nonlinear dynamics with implicit relative degree, there are two main challenges. Firstly, under the condition of only local interaction, each follower needs to obtain the high-order derivative information of the reference signal and apply it to the controller design. Secondly, the implicit system structure makes it impossible to carry out backstepping design. To achieve output consensus in such MASs, the two aforementioned issues are addressed as follows. For the first issue, a novel filter is designed to indirectly obtain the high-order derivatives of the observation signals. For the second issue, a recurrent high-order neural network (RHONN) is employed to approximate the original system, and an augmented error-based stability analysis method is adopted. By combining the new filter with the augmented error-based analysis method, we propose an effective control scheme and a complete analysis process for the cooperative problem of multiple noncanonical nonlinear systems, with the results guaranteeing the boundedness of all closed-loop signals and the asymptotic output consensus tracking. These theoretical findings are well verified by simulation tests.

1. Introduction

Over the past few decades, tremendous efforts have been devoted to multi-agent systems (MASs) due to their widespread applications in practical engineering (Burmeister et al., 1997; Nagata & Sasaki, 2002; Nie et al., 2025; Yang et al., 2024). The cooperative control problem of MASs plays an important role in fields such as traffic networks (Zhonghe et al., 2015), mobile robots (Ao & Luo, 2024; Liang et al., 2023), microgrids (Yang et al., 2018), and hardware clock systems (Wang et al., 2024). When studying this problem, one persistent issue cannot be avoided, which concerns how to realize global agreement only through local interactions. Such a problem, however, is not considered adequately in some earlier studies, e.g., Junjie and Wang (2013) and Zhang et al. (2016a), in which it is straightforwardly assumed that each follower can acquire leader-related information without direct connection to the leader. To address this problem, a distributed observer-based strategy is proposed in Su and Huang (2012), and based on this strategy, the observation value of reference signal can be transmitted to each follower through local communication between agents. Such a general observer, defined by a first-order differential equation, is often used to solve the cooperative output regulation problem of linear systems,

where the control process only utilizes the observed signal itself rather than its derivative form Cai et al. (2017), He and Huang (2022), Su and Huang (2012). Therefore, for control schemes where the control process requires the participation of high-order derivatives of reference signals, such as backstepping method or the control approach in this paper, general observers are not applicable. The fundamental reason is that when calculating the second-order or higher-order derivatives of the observer of the current agent, this process involves information from the neighbors of neighbors, which clearly violates the condition of local interaction. To this end, researchers have designed high-order observers (Li & Tong, 2023; Liu et al., 2024a; Tan et al., 2024; Zhang et al., 2023a) using a chain structure in recent years. These high-order observers address the shortcomings of general observers, but still have limitations.

To facilitate the analysis of their limitations, a brief overview of these works is first provided. In Zhang et al. (2023a), a resilient distributed observer (RDO) is developed to address the cooperative output regulation issue using backstepping techniques. In Li and Tong (2023), a circular switching filter (CSF) is proposed, which also addresses the cooperative control problem of nonlinear MASs by integrating backstepping techniques. In Liu et al. (2024a), a novel low-pass filter (LPF) is designed and used to handle the cooperative resilient tracking problem of a class

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of nonlinear MASs. Regarding the three methods of RDO, CSF, and LPF, although all can solve the high-order derivative problem, each has certain limitations. Specifically, in the stability analysis of the observer, the RDO in Zhang et al. (2023a) requires the observability of a specific matrix pair; whereas the CSF in Li and Tong (2023) necessitates that all eigenvalues of the leader system matrix be semi-simple with zero real parts. In applications, the LPF in Liu et al. (2024a) requires the leader's system matrix to be globally known in order to achieve high-order differentiation under local interaction conditions. Besides the aforementioned limitations, a common limitation shared by Li and Tong (2023), Zhang et al. (2023a) and Liu et al. (2024a) is the requirement for the system matrix to be known, which is a notable constraint. So, there is still much room for improvement in these high-order observers. More recently, Tan et al. (2024) addressed the common limitation of requiring global knowledge of the leader's system matrix by adaptively estimating it. Nonetheless, as its estimation is still presented in the form of a first-order differential equation, it cannot be well applied to the control design in this paper. Therefore, given the limitations of previous studies, developing a more universal observer is essential.

In addition to the problem of information transmission, we also need to consider what analytical method should be adopted for the system studied in this paper. Until now, the majority of studies on MASs consensus control still focus on linear systems (Chen et al., 2023; Liu et al., 2024b; Tan & Wang, 2024) or canonical nonlinear systems with explicit relative degree (Li et al., 2024, 2019; Luo et al., 2024; Lv et al., 2025; Tan et al., 2026; Zhang et al., 2023b). For these forms of systems, we can directly design controllers and conduct stability analysis using the Lyapunov-based backstepping method. However, in practical applications, nonlinear systems are not often in canonical form. A typical example is a MAS composed of nine-state variable aircraft (Tao, 2014), where each agent is modeled as a noncanonical nonlinear system. For such MASs, where the relative degree of each agent is implicit and has large parametric uncertainties, the design based on the backstepping framework becomes ineffective. Addressing the stability analysis of these systems, therefore, necessitates novel strategies. To the best of our knowledge, the combination of the observer with the gradient-based adaptive control method (Lai et al., 2020) will be a better solution.

In order to address the consensus tracking problem of MASs where the followers exhibit noncanonical nonlinear dynamics, this article designs a more applicable adaptive distributed observer and proposes an effective control scheme based on it. In this study, firstly, a recurrent high-order neural network (RHONN) is adopted to approximate the follower systems, making all terms parametrizable. Secondly, an extended feedback linearization technique is employed to define the relative degree of the follower systems. Finally, a control scheme is designed along with the observer, and different analytical methods are applied based on scenarios of different relative degrees. In conclusion, the novel contributions of this work are described as follows.

- (1) Existing control methods for noncanonical nonlinear systems (Lai et al., 2020, 2022) are predominantly developed for single system and rely on augmented-error-based analysis. This paper extends such control framework to the multi-agent cooperative domain. By integrating a RHONN model with an adaptive distributed observer, an effective distributed control scheme is proposed that achieves asymptotic output consensus tracking for noncanonical nonlinear multi-agent systems.
- (2) Since standard adaptive distributed observers (Cai et al., 2017; He & Huang, 2022; Su & Huang, 2012) only supply leader state and first-derivative information, they are insufficient for the noncanonical nonlinear setting considered here. We design a filter that provides each follower with estimates of the leader's higher-order derivatives, overcoming the limitations of prior works and enabling gradient-based adaptive control to be applied to consensus problems for this wider class of systems.

- (3) Most existing consensus methods for nonlinear MASs (Chen et al., 2023; Li et al., 2024, 2019; Liu et al., 2024b; Luo et al., 2024; Lv et al., 2025; Tan & Wang, 2024; Tan et al., 2026; Zhang et al., 2023b) either guarantee only bounded-error tracking, or achieve asymptotic convergence under the assumption that the system relative degree is exactly known a priori. In this paper, by employing the RHONN model together with an extended feedback linearization technique, distinct control schemes are developed for different relative degree cases without requiring the relative degree to be specified in advance. Moreover, the stability analysis, based on augmented error dynamics for the relative-degree-two case, is presented in a manner that naturally extends to any higher relative degree scenario.

The organization of this paper is as follows. Section 2 presents preliminary knowledge, the MASs model, and the consensus tracking control problem. Section 3 first establishes the approximation accuracy of the RHONN model, and then derives the relative degree conditions of the studied MASs via the extended feedback linearization technique. Section 4 designs a novel filter to ensure the existence of high-order derivatives for the general adaptive distributed observer. Based on the derived conditions and proposed filter, Section 5 presents adaptive control schemes and stability analysis methods tailored to systems with different relative degrees. Section 6 conducts detailed simulations to verify the proposed results. Finally, conclusions are drawn in Section 7.

2. Preliminary knowledge and problem statement

2.1. Notations and graph theory

For a vector $x \in \mathbb{R}^n$ and a square matrix $M \in \mathbb{R}^{n \times n}$, $\|x\|$ denotes the Euclidean norm of x , while $\sigma(M)$ represents the spectrum of M . Additionally, $\Re(\sigma(M))$ represents the set of real parts of $\sigma(M)$; $\bar{\delta}_M = \max(\Re(\sigma(M)))$, and $\underline{\delta}_M = \min(\Re(\sigma(M)))$.

In this study, a static communication network composed of $(N + 1)$ agents with N followers and a unique leader is defined as a diagraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, while its leader-excluding undirected subgraph is denoted as $\mathcal{G}_s = (\mathcal{V}_s, \mathcal{E}_s)$. Here, $\mathcal{V} = \{v_0, v_1, \dots, v_N\}$ is the vertex set and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the edge set, and $\mathcal{V}_s$ and $\mathcal{E}_s$ are sets that exclude the leader v_0 . Additionally, we define the neighbor set of agent i as $\mathcal{N}_i = \{j \mid (j, i) \in \mathcal{E}\}$ and define the adjacency matrix of $\mathcal{G}_s$ as $A = [a_{ij}] \in \mathbb{R}^{N \times N}$, where $a_{ij} > 0$ if $j \in \mathcal{N}_i$, and $a_{ij} = 0$ otherwise. The Laplacian matrix $L = [L_{ij}] \in \mathbb{R}^{N \times N}$ of $\mathcal{G}_s$ is constructed as $L_{ii} = \sum_{j \neq i} a_{ij}$ and $L_{ij} = -a_{ij}$, $i \neq j$. A diagonal matrix $P = \text{diag}[\mu_i] \in \mathbb{R}^{N \times N}$ represents follower-leader affiliation: node i observes the leader if $\mu_i > 0$, otherwise it is zero.

Assumption 1. In the graph $\mathcal{G}$, there exists a spanning tree rooted at a unique leader.

2.2. Systems description and problem formulation

The leader's dynamics are considered to be governed by the following equations:

$$\dot{x}_0(t) = S_0 x_0(t), \quad y_0(t) = C_0 x_0(t), \quad (1)$$

where $x_0(t) \in \mathbb{R}^n$ is a full-state vector, $y_0(t) \in \mathbb{R}$ is the leader-system output, which indeed acts as the reference signal $y_d(t)$ of the follower systems, and $S_0 \in \mathbb{R}^{n \times n}$ and $C_0^T \in \mathbb{R}^n$ represent a constant matrix and a constant vector, respectively.

Assumption 2. The matrix S_0 has no eigenvalues with positive real parts.

In the subgraph $\mathcal{G}_s$, each follower is modeled as a single-input and single-output (SISO) noncanonical nonlinear system, as given below

$$\dot{x}_i^o(t) = f_i^o(x_i^o(t)) + B_i u_i(t), \quad y_i^o(t) = C_i x_i^o(t) \quad (2)$$

where $x_i^o(t) \in \mathbb{R}^n$ is the system state vector, $y_i^o(t) \in \mathbb{R}$ is the output signal, $u_i(t) \in \mathbb{R}$ is the input signal; B_i and C_i^T are unknown constant vectors

in $\mathbb{R}^n$, and $f_i^o(x_i^o) : \Omega \rightarrow \mathbb{R}^n$ represents an unknown Lipschitz continuous function defined over a compact set Ω .

Remark 1. A noncanonical nonlinear system refers to a nonlinear system that does not have the predefined structure of a canonical system, such as an explicit relative degree, and its approximate model also possesses this characteristic. Due to this feature of its dynamic model, the Lyapunov-based backstepping method cannot be used for control design. Meanwhile, since the parameters B_i , C_i , and the function $f_i^o(x_i^o)$ in the system are all unknown, feedback linearization cannot be employed for further design either. $\diamond$

Since the nonlinear function $f_i^o(x_i^o)$ in follower model (2) is unknown and unparametrizable, we adopt an RHONN to remodel the original system similarly to our previous works (Lai et al., 2020, 2022), so that all terms are parametrizable completely. On this basis, the following RHONN-based approximation system model can be obtained:

$$\begin{aligned}\dot{x}_i(t) &= A_i x_i(t) + W_i^* S_i(x_i(t)) + B_i u_i(t) \\ y_i(t) &= C_i x_i(t).\end{aligned}\quad (3)$$

where $A_i \in \mathbb{R}^{n \times n}$ denotes a constant Hurwitz matrix, $W_i^* \in \mathbb{R}^{n \times l}$ represents an unknown ideal weight matrix, $S_i(x_i(t)) = [S_{i,1}(x_i(t)), S_{i,2}(x_i(t)), \dots, S_{i,l}(x_i(t))]^T \in \mathbb{R}^l$, and the elements $S_{i,j}(x_i(t))$ represent the activation function, which means the response of the j th neuron to the input x_i . For the activation function, its form is given by:

$$S_{i,j}(x_i) = \prod_{k \in I_j} (s_i(x_{i,k}))^{d_{k,j}} \quad (4)$$

with

$$s_i(x_{i,k}) = \frac{\alpha_k}{1 + e^{-\beta_k x_{i,k}}} + \gamma_k, \quad (5)$$

where I_j , $j = 1, 2, \dots, l$ being subsets of the collection $\{1, 2, \dots, n\}$, $d_{k,j} \geq 0$, $j = 1, 2, \dots, l$ are integers, and the function $s_i(x_{i,k})$ is a sigmoid function with adjustable parameters α_k , β_k , γ_k being constant satisfying $\alpha_k, \beta_k \in \mathbb{R}^+$, and $\gamma_k \in \mathbb{R}$.

Control objective. Given a class of MASs composed of N followers modeled by (3) and a leader modeled by (1), this study aims to design a distributed control law $u_i(t)$, such that the output of each follower asymptotically synchronizes to the leader's output (i.e. $y_i(t) \rightarrow y_0(t)$ as $t \rightarrow \infty$ for all $i = 1, 2, \dots, N$), with all closed-loop signals remaining bounded.

3. Performance and relative degree analysis of approximate models

In this section, we first establish the approximation capability of the RHONN model, showing that it can approximate the original noncanonical nonlinear system to any desired degree of accuracy. Based on this, we then employ the extended feedback linearization technique to define the relative degree of the RHONN-based approximation system for the subsequent control design.

3.1. Approximation accuracy of the RHONN model

Since the original nonlinear dynamics $f_i^o(x_i^o)$ are unknown and unparameterized, we employ an RHONN to construct a parameterizable approximation model in the form of (3). Before proceeding to the control design, we first clarify the approximation accuracy of the RHONN model, which provides the theoretical basis for the subsequent analysis.

In a complete description, the true system dynamics incorporating the neural network approximation error can be written as

$$\dot{x}_i = A_i x_i + W_i^* S_i(x_i) + B_i u_i + \delta_i^a(t), \quad (6)$$

where $\delta_i^a(t)$ denotes the residual approximation error. As established in Kosmatopoulos et al. (1995), such an error mainly arises from the limited number of high-order connections in the RHONN structure, and it

can be made arbitrarily small over any compact operating set by increasing the number of high-order terms. This property is formally stated as follows.

Proposition 1. Suppose that the original system (2) and the RHONN model (3) share the same initial state, i.e., $x_i^o(0) = x_i(0)$. Then, for any given precision $\varsigma > 0$ and any finite time $T > 0$, there exist a sufficiently large number of high-order connections l and an ideal weight matrix $W_i^* \in \mathbb{R}^{n \times l}$ such that the state $x_i(t)$ of the RHONN model satisfies

$$\sup_{0 \leq t \leq T} \|x_i(t) - x_i^o(t)\| \leq \varsigma.$$

Proof. See Appendix A. $\square$

Based on the above approximation property, we omit RHONN approximation errors in our main derivations for simplicity, to fully focus on cooperative control of MASs modeled via noncanonical RHONN-based dynamics. This ideal setup assumes the true dynamics (2) are exactly captured by RHONN with optimal weights. If modeling errors or disturbances are included, extra robust compensation laws must be designed. Our error-free analysis lays a baseline for future robust extensions, which we leave for follow-up research.

3.2. Conclusions related to feedback linearization

Consider the follower system model given by (3). Setting $f_i(x_i) = A_i x_i + W_i^* S_i(x_i)$, $g_i(x_i) = B_i$, and $h_i(x_i) = C_i x_i$, we can easily derive that

$$L_{g_i} h_i(x_i) = C_i B_i \quad (7)$$

$$L_{g_i} L_{f_i} h_i(x_i) = C_i A_i B_i + C_i W_i^* \frac{\partial S_i(x_i)}{\partial x_i} B_i,$$

where $L_{(\cdot)}(\cdot)$ represents the Lie derivative.

Definition 1 (See Sastry & Bodson, 1989). Within the compact set $\Omega \subseteq \mathbb{R}^n$, the relative degree of the nonlinear system specified by (3) is ρ , if for all $x_i \in \mathbb{R}^n$

$$L_{g_i} L_{f_i}^k h_i(x_i) = 0, \quad k = 0, 1, \dots, \rho - 2 \quad (8)$$

$$L_{g_i} L_{f_i}^{\rho-1} h_i(x_i) \neq 0.$$

Lemma 1 (See Zhang et al., 2016b). Provided that the relative degree of system (3) on compact set Ω is ρ , by means of diffeomorphism $[\xi_i^T(t), \eta_{i,z}^T(t)]^T = T(x_i) = [T_c^T(x_i(t)), T_z^T(x_i(t))]^T \in \mathbb{R}^n$ with $\xi_i(t) = T_c(x_i) = [\xi_{i,1}(t), \xi_{i,2}(t), \dots, \xi_{i,\rho}(t)]^T = [h_i(x_i), L_{f_i} h_i(x_i), \dots, L_{f_i}^{\rho-1} h_i(x_i)]^T \in \mathbb{R}^\rho$ and $\eta_{i,z}(t) = T_z(x_i) \in \mathbb{R}^{n-\rho}$, it is possible to transform the system into two subsystems, namely the tracking control dynamics subsystem (TCDS)

$$\begin{aligned}\dot{\xi}_{i,j}(t) &= \xi_{i,j+1}(t), \quad j = 1, 2, \dots, \rho - 1 \\ \dot{\xi}_{i,\rho}(t) &= L_{f_i}^\rho h_i(x_i) + L_{g_i} L_{f_i}^{\rho-1} h_i(x_i) u_i(t)\end{aligned} \quad (9)$$

and the zero dynamics subsystem (ZDS)

$$\dot{\eta}_{i,z}(t) = \Psi(\xi_i, \eta_{i,z}) \quad (10)$$

where Ψ is a nonlinear function of ξ_i and $\eta_{i,z}$.

3.3. Relative degree conditions of the followers

Under Definition 1, the following lemma is provided for the different relative degrees relevant to this article.

Lemma 2 (See Zhang et al., 2016b). (i) On the compact set Ω , system (3) possesses a relative degree one if and only if

$$C_i B_i \neq 0. \quad (11)$$

(ii) On the compact set Ω , system (3) possesses a relative degree two if and only if

$$C_i B_i = 0 \text{ and } C_i A_i B_i + C_i W_i^* \frac{\partial S_i(x_i)}{\partial x_i} B_i \neq 0 \quad \forall x \in \Omega. \quad (12)$$

Remark 2. Within this study, we exclude scenarios where system (3) has a relative degree exceeding two, for the following two reasons. Firstly, according to Definition 1, if the relative degree is greater than two, then

$$C_i A_i B_i + C_i W_i^* \frac{\partial S_i(x_i)}{\partial x_i} B_i = 0 \quad (13)$$

needs to be satisfied for $\forall x \in \Omega$. However, due to the influence of W_i^* and $\partial S(x_i)/\partial x_i$, this is difficult to achieve. Secondly, from the subsequent derivation, it is not difficult to find that the research on follower systems (3) with a relative degree of two serves as an example that can be generalized to systems of any degree. $\diamond$

4. Adaptive distributed observer and its high-order form

This section will first introduce a general adaptive distributed observer, and then presents a new filter to address the problem of non-existent high-order derivatives in this observer.

4.1. Adaptive distributed observer

The adaptive distributed observer in Cai and Huang (2021) has the following form:

$$\dot{\hat{S}}_i(t) = \mu_1 \sum_{j=0}^N a_{ij} (\hat{S}_j(t) - \hat{S}_i(t)) \quad (14)$$

$$\dot{\hat{C}}_{i,0}(t) = \mu_2 \sum_{j=0}^N a_{ij} (\hat{C}_{j,0}(t) - \hat{C}_{i,0}(t)) \quad (15)$$

$$\dot{\eta}_{i,0}(t) = \hat{S}_i(t) \eta_{i,0}(t) + \mu_3 \sum_{j=0}^N a_{ij} (\eta_{j,0}(t) - \eta_{i,0}(t)) \quad (16)$$

where $\hat{S}_i(t)$, $\hat{C}_{i,0}(t)$ and $\eta_{i,0}(t)$ are the estimates of S_0 , C_0 and x_0 respectively, with $\hat{S}_0 = S_0$, $\hat{C}_{0,0} = C_0$, $\eta_{0,0} = x_0$ and $\mu_1, \mu_2, \mu_3 > 0$. For this observer, the description is given in Cai and Huang (2021, Lemma 2.3) as follows.

Lemma 3. Under Assumption 1, systems (1) and (14)–(16) satisfy the following properties for any initial conditions $\hat{S}_i(0)$, $\eta_{i,0}(0)$, $\hat{C}_{i,0}(0)$ with $i = 1, 2, \dots, N$, and $x_0(0)$.

- (i) Given any $\mu_1, \mu_2 > 0$, $\lim_{t \rightarrow \infty} (\hat{S}_i(t) - S_0) = 0$, $\lim_{t \rightarrow \infty} (\hat{C}_{i,0}(t) - C_0) = 0$ exponentially $\forall t \geq 0$ at least at the rate of $\mu_1 \underline{\delta}_L$, $\mu_2 \underline{\delta}_L$, respectively;
- (ii) Given $\mu_1, \mu_2, \mu_3 > \bar{\delta}_{S_0} \underline{\delta}_L^{-1}$, $\lim_{t \rightarrow \infty} (\eta_{i,0}(t) - x_0(t)) = 0$ exponentially for all $t \geq 0$.

A close observation of observer (14)–(16) reveals that its second-order and higher-order derivatives cannot be directly computed via local interaction. Specifically, when we take the observed state $\eta_{i,0}(t)$ in Eq. (16) as an example and calculate its second-order derivative, the right-hand side of the equation contains the term $\dot{\eta}_{j,0}(t)$. However, for the i -th agent, obtaining $\dot{\eta}_{j,0}$ requires information about its neighbors, i.e., $\mathcal{N}_j$, which clearly violates the local interaction condition. Therefore, to address the issue of acquiring high-order derivatives for observer (14)–(16), we will design a new filter in the following section.

Remark 3. Since the control objective requires all closed-loop signals to be bounded, it is necessary here to explain the boundedness of the observation signals $\hat{S}_i(t)$, $\hat{C}_{i,0}(t)$ and $\eta_{i,0}(t)$. According to Lemma 3, we know that the observation signals can exponentially converge to the target signals, and S_0 , C_0 and x_0 are all bounded. Therefore, the boundedness of $\hat{S}_i(t)$, $\hat{C}_{i,0}(t)$ and $\eta_{i,0}(t)$ can be easily inferred. $\diamond$

4.2. High-order adaptive distributed observer

Since the second derivatives of the observation signals $\hat{S}_i(t)$, $\hat{C}_{i,0}(t)$ and $\eta_{i,0}(t)$ do not exist, we design auxiliary variables to approximate these signals, thereby avoiding direct high-order differentiation of them.

For this purpose, the following key lemma is presented.

Lemma 4. Considering the following system:

$$\begin{aligned} \dot{y}(t) = & -a(y(t) - x(t)) - d(t) \left(\tanh \frac{\varepsilon(t)}{\rho(t) + v} \right. \\ & \left. - \bar{v} - v \tanh \frac{\varepsilon(t)}{v} \right) \end{aligned} \quad (17)$$

$$\dot{d}(t) = \lambda \varepsilon^T(t) \left(\tanh \frac{\varepsilon(t)}{\rho(t) + v} + \bar{v} + v \tanh \frac{\varepsilon(t)}{v} \right),$$

where a , λ , and v are constants to be designed, $\bar{v} = [v, v, \dots, v]^T$, $x(t)$, $y(t) \in \mathbb{R}^q$, $d(t)$ is an estimate of $d^* = \|\dot{x}(t)\|_\infty$ with $\dot{x}(t)$ being bounded, $\varepsilon = y(t) - x(t)$ indicates the error between $y(t)$ and $x(t)$, $\rho(t)$ is a first-order differentiable positive L^1 function such that $\rho(t)$ and $\dot{\rho}(t)$ are bounded, $\tanh \frac{\varepsilon(t)}{\rho(t) + v} = [\tanh \frac{\varepsilon_1(t)}{\rho(t) + v}, \tanh \frac{\varepsilon_2(t)}{\rho(t) + v}, \dots, \tanh \frac{\varepsilon_n(t)}{\rho(t) + v}]^T$ is a function vector. If the derivatives of $x(t)$ and $\dot{x}(t)$ are bounded, then:

- (i) $\lim_{t \rightarrow \infty} (y(t) - x(t)) = 0$ asymptotically;
- (ii) $y(t)$, $\dot{y}(t)$, and $\ddot{y}(t)$ are bounded.

Proof. First, the chosen Lyapunov function is

$$V(\varepsilon, \bar{d}) = \frac{1}{2} \varepsilon^T(t) \varepsilon(t) + \frac{1}{2\lambda} \bar{d}^2(t). \quad (18)$$

Here, $\bar{d}(t)$, which represents the error in the estimation process, is defined as $\bar{d}(t) = d(t) - d^*$. Subsequently, substituting (17) into the time derivative of (18) yields

$$\dot{V} \leq -a \|\varepsilon(t)\|^2 + d^* \sum_{i=1}^q 0.2785 (\rho(t) + v + v^2). \quad (19)$$

When $v = -1$, we can obtain

$$\dot{V} \leq -a \|\varepsilon(t)\|^2 + d^* \sum_{i=1}^q 0.2785 \rho(t). \quad (20)$$

Performing integration on each side of (20) simultaneously on the interval $[0, t]$ leads to

$$V(t) \leq V(0) - a \int_0^t \|\varepsilon(\tau)\|^2 d\tau + d^* \sum_{i=1}^q 0.2785 \int_0^t \rho(\tau) d\tau. \quad (21)$$

In the inequality (21), function $\rho(t) \in L^1$, therefore $V(t) \in L^\infty$, which suggests form (18) that $\varepsilon(t)$, $\bar{d}(t) \in L^\infty$. Thus, $y(t)$ and $d(t)$ are bounded. Considering both (17) and (21), it is known that $\dot{y}(t)$ and $\dot{d}(t)$ are bounded due to (17), and we can infer that $\|\varepsilon(t)\|$ satisfies the properties of uniformly continuous and square integrable. Then, $\|\varepsilon(t)\|$ will converge to zero as time approaches infinity, according to Barbalat's lemma.

Regarding the boundedness of $\ddot{y}(t)$, according to (17), since both $\dot{y}$ and $\dot{x}(t)$ are bounded, it suffices to verify the boundedness of the derivative of $\Lambda(t) = d(t) \left(\tanh \frac{\varepsilon(t)}{\rho(t) + v} - \bar{v} - v \tanh \frac{\varepsilon(t)}{v} \right)$. Obviously,

$$\begin{aligned} \dot{\Lambda}(t) = & \dot{d}(t) \left(\tanh \frac{\varepsilon(t)}{\rho(t) + v} - \bar{v} - v \tanh \frac{\varepsilon(t)}{v} \right) \\ & + d(t) \left(\operatorname{sech}^2 \left(\frac{\varepsilon(t)}{\rho(t) + v} \right) \cdot \frac{\dot{\varepsilon}(t)(\rho(t) + v) - \varepsilon(t)\dot{\rho}(t)}{(\rho(t) + v)^2} \right. \\ & \left. - \operatorname{sech}^2 \left(\frac{\varepsilon(t)}{v} \right) \cdot \dot{\varepsilon}(t) \right), \end{aligned} \quad (22)$$

where each term is bounded. Thus, $\ddot{y}(t)$ is bounded. $\square \square$

Remark 4. Lemma 4 avoids direct differentiation of x by designing an auxiliary variable y that converges to x . When higher-order derivatives of x are needed, a corresponding number of auxiliary variables can be designed such that $y_1 \rightarrow y_2 \rightarrow \dots \rightarrow x$, thereby meeting the requirement. $\diamond$

Based on Lemma 4, to design a distributed adaptive observer capable of calculating the second derivative using only neighbor information, this paper first designs two dynamic auxiliary systems for the variables

$\hat{C}_{i,0}(t)$ and $\eta_{i,0}(t)$ that require second-order differentiation, respectively. Specifically, the dynamic auxiliary system for $\hat{C}_{i,0}(t)$ is given as

$$\begin{aligned} \dot{\hat{C}}_{i,1}(t) = & -a_i(\hat{C}_{i,1}(t) - \hat{C}_{i,0}^T(t)) \\ & - m_i(t) \left(\tanh \frac{\varepsilon_{ci}(t)}{\varrho_i(t) + \nu} - \bar{\nu} - \nu \tanh \frac{\varepsilon_{ci}(t)}{\nu} \right) \end{aligned} \quad (23)$$

$$\dot{m}_i(t) = \lambda \varepsilon_{ci}^T(t) \left(\tanh \frac{\varepsilon_{ci}(t)}{\varrho_i(t) + \nu} + \bar{\nu} + \nu \tanh \frac{\varepsilon_{ci}(t)}{\nu} \right) \quad (24)$$

where $\varepsilon_{ci}(t) = \hat{C}_{i,1}(t) - \hat{C}_{i,0}^T(t)$ is the tracking error, $m_i(t)$ is the estimation of $m_i^* = \|\dot{\hat{C}}_{i,0}(t)\|_\infty$. The dynamic auxiliary system for $\eta_{i,0}(t)$ is expressed as

$$\begin{aligned} \dot{\eta}_{i,1}(t) = & -b_i(\eta_{i,1}(t) - \eta_{i,0}(t)) \\ & - d_i(t) \left(\tanh \frac{\varepsilon_{\eta i}(t)}{\varrho_i(t) + \nu} - \bar{\nu} - \nu \tanh \frac{\varepsilon_{\eta i}(t)}{\nu} \right) \end{aligned} \quad (25)$$

$$\dot{d}_i(t) = \lambda \varepsilon_{\eta i}^T(t) \left(\tanh \frac{\varepsilon_{\eta i}(t)}{\varrho_i(t) + \nu} + \bar{\nu} + \nu \tanh \frac{\varepsilon_{\eta i}(t)}{\nu} \right) \quad (26)$$

where $\varepsilon_{\eta i}(t) = \hat{\eta}_{i,1}(t) - \hat{\eta}_{i,0}(t)$ is the tracking error, $d_i(t)$ is the estimation of $d_i^* = \|\dot{\hat{\eta}}_{i,0}(t)\|_\infty$.

Combining systems (14)–(16) and (17), this paper designs a method for obtaining the high-order derivatives of the observation signals. Realized via the auxiliary systems (23)–(26), the relevant lemma is presented as follows by synthesizing Lemmas 3 and 4.

Lemma 5. *Given the leader system (1), the adaptive distributed observer (14)–(16), and the auxiliary systems (23)–(26), under Assumption 1, for any initial conditions $\hat{S}_i(0)$, $\eta_{i,0}(0)$, $\hat{C}_{i,0}(0)$ with $i = 1, 2, \dots, N$, and $x_0(0)$, the following results hold:*

- (i) for any $\mu_1, \mu_2 > 0$, $\lim_{t \rightarrow \infty} (\hat{S}_i(t) - S) = 0$ exponentially for all $t \geq 0$ at least at the rate of $\mu_1 \underline{\delta}_L$, $\lim_{t \rightarrow \infty} (\hat{C}_{i,1}^T(t) - C_0) = 0$ asymptotically;
- (ii) when $\mu_1, \mu_2, \mu_3 > \bar{\delta}_{S_0} \bar{\delta}_L^{-1}$, $\lim_{t \rightarrow \infty} (\eta_{i,1}(t) - x_0(t)) = 0$ asymptotically, further $\lim_{t \rightarrow \infty} (\hat{C}_{i,1}^T(t) \eta_{i,1}(t) - y_0(t)) = 0$.

Remark 5. Lemma 5 only indicates that the auxiliary variables $\hat{C}_{i,1}$ and $\eta_{i,1}$ can converge to the corresponding observation signals, but it does not show that the derivatives of the auxiliary variables can also approach the derivative information of the leader. This is actually reasonable. As analyzed in Section 5.2, to construct the second-order differential equation as shown in (39) and thereby achieve output tracking, we only need to ensure the tracking performance of $y_{di} = \hat{C}_{i,1} \eta_{i,1}$ with respect to y_0 , and that the high-order derivatives of y_{di} are mathematically accessible. $\diamond$

5. Adaptive consensus control design of approximation models

For follower systems (3) with relative degree not exceeding two, in this section, a distributed controller is designed using an adaptive approach to achieve consensus tracking and ensure the closed-loop stability.

In the process of further derivation, a stable zero dynamics subsystem is essential for designing a stable distributed output tracking control scheme. Therefore, we have made the assumption below concerning input-to-state stability (ISS).

Assumption 3. The ZDS $\dot{\eta}_{i,z} = \Psi(\xi_i, \eta_{i,z})$ satisfies the inequality below, being ISS relative to ξ_i as the input:

$$\|\eta_{i,z}(t)\| \leq \varsigma(\|\eta_{i,z}(t_0)\|, t - t_0) + \varpi \left(\sup_{t_0 \leq \tau \leq t} \|\xi_i(\tau)\| \right) \quad (27)$$

where for each given a , $\varsigma(a, b)$ is a decreasing function of b , approaching zero as $b \rightarrow 0$. For each given b , $\varsigma(a, b)$ increases strictly with a , and $\varsigma(0, b) = 0$. In addition, $\varpi(b)$ is strictly increasing and fulfills $\varpi(0) = 0$.

Remark 6. Given inequality (27), we can easily infer from the properties of $\varsigma(\cdot)$ and $\varpi(\cdot)$ that as t increases, if ξ_i is bounded, then

$\eta_{i,z}$ is bounded. Meanwhile, based on the change of coordinates $[\xi_i^T(t), \eta_{i,z}^T(t)]^T = T(x_i)$, $x_i(t)$ and $[\xi_i^T(t), \eta_{i,z}^T(t)]^T$ have mutually dependent boundedness. Hence, under Assumption 3, when $\xi_i(t)$ is bounded, so is $\eta_{i,z}(t)$, and subsequently, $x_i(t)$ is also bounded, with the reverse also being valid. $\diamond$

In what follows, adaptive control schemes are designed for different relative degrees, considering unknown follower system parameters and permitted local interactions.

5.1. Control strategies for systems with $\rho = 1$

For the follower system (3) with $\rho = 1$, in order to design a distributed controller to achieve consensus within the constraint of only being able to obtain neighbor information, an ADO is used in this section.

In case where the parameters B_i , C_i , and W_i^* of each follower are unspecified, the design process of the distributed control scheme is as follows.

5.1.1. System model

According to Lemma 1, the TCDS of each follower can be described as

$$\dot{y}_i = C_i A_i x_i(t) + C_i B_i u_i(t) + C_i W_i^* S_i(x_i(t)) \quad (28)$$

where $C_i B_i \neq 0$. With some manipulation on (28), the parametrized expression

$$\dot{y}_i(t) = \theta_i^{*T} \phi_i(t), \quad i = 1, \dots, N \quad (29)$$

is derived, where $\theta_i^* = [C_i, C_i W_i^*, C_i B_i]^T \in \mathbb{R}^{n+l+1}$ represents a vector with unknown constant elements, while $\phi_i(t) = [(A_i x_i)^T, S_i^T, u_i]^T \in \mathbb{R}^{n+l+1}$ denotes a vector with known elements.

5.1.2. Adaptive controller structure

Based on (29), the distributed controller is developed as

$$\begin{aligned} u_i(t) = & \frac{1}{k_{i,3}(t)} \left(-k_{i,1}^T(t) A_i x_i(t) - k_{i,2}^T(t) S_i(x_i(t)) \right. \\ & \left. + \hat{C}_{i,0}(t) \eta_{i,0}(t) + \hat{C}_{i,0}(t) \dot{\eta}_{i,0}(t) - a e_i(t) \right) \end{aligned} \quad (30)$$

where $a > 0$ is a constant, $k_{i,1}(t) \in \mathbb{R}^n$, $k_{i,2}(t) \in \mathbb{R}^l$, and $k_{i,3}(t) \in \mathbb{R}$ serve as respective estimates for C_i^T , $(C_i W_i^*)^T$, and $C_i B_i$, and $e_i(t) = y_i(t) - y_{di}(t)$ with $y_{di}(t) = \hat{C}_{i,0} \eta_{i,0}$.

Remark 7. Since the leader's signal $y_0(t)$ is not globally available, the observation signal $y_{di}(t)$ is adopted to replace the reference signal provided by the leader (1).

5.1.3. Adaptive law

For the estimation $\theta_i(t)$, we devise the adaptive law

$$\dot{\theta}_i(t) = \Gamma e_i(t) \phi_i(t). \quad (31)$$

In the above-mentioned equation, Γ is a diagonal matrix represented as $\Gamma = \text{diag}\{\gamma_1, \dots, \gamma_{n+l+1}\}$, where $\gamma_i > 0$ holds for $i = 1, \dots, n+l+1$. Additionally, $\theta_i(t)$ is defined as $\theta_i(t) = [k_{i,1}^T(t), k_{i,2}^T(t), k_{i,3}(t)]^T$.

5.1.4. Stability analysis

Some vectors are defined as follows. Denote $\tilde{\theta}_i(t)$ as the difference between $\theta_i(t)$ and θ_i^* , i.e., $\tilde{\theta}_i(t) = \theta_i(t) - \theta_i^*$. Let $\theta(t) = [\theta_1^T(t), \theta_2^T(t), \dots, \theta_N^T(t)]^T$, $\theta^* = [\theta_1^{*T}, \theta_2^{*T}, \dots, \theta_N^{*T}]^T$, $y(t) = [y_1(t), y_2(t), \dots, y_N(t)]^T$, and $e(t) = [e_1(t), e_2(t), \dots, e_N(t)]^T$ be the concatenated vector. Subsequently, the Lyapunov function is defined as

$$V(e, \tilde{\theta}) = \frac{1}{2} e^T(t) e(t) + \frac{1}{2} \tilde{\theta}^T(t) \Gamma^{-1} \tilde{\theta}(t), \quad (32)$$

where $\Gamma = I_N \otimes \Gamma$. Then, taking the time derivative of (32) yields

$$\dot{V}(e, \tilde{\theta}) = \sum_{i=1}^N -ae_i^2(t) \leq 0. \quad (33)$$

Consequently, $e(t)$ and $\tilde{\theta}(t)$ are bounded, thereby implying the boundedness of $y(t)$ and $\theta(t)$. By [Assumption 3](#), the boundedness of $\xi_i(t) = y_i(t)$ ensures bounded $x_i(t)$. Using (30) and [Lemma 3](#), u_i is bounded, so is $\phi_i(t)$. With $\sum_{i=1}^N \int_0^\infty e_i^2(t)dt$ finite and $\dot{e}(t)$ remaining bounded, by Barbalat's lemma, we have $\lim_{t \rightarrow \infty} e_i(t) = 0$. Therefore, the following conclusions can be drawn.

Theorem 1. *Considering [Assumptions 1-3](#), when the adaptive distributed observer (14)–(16) and the controller (30) are implemented on the system (3) with $\rho = 1$ and unknown parameters B_i, C_i , and W_i^* , it can be ensured that every signal within the closed-loop system remains bounded. Moreover, the tracking error $e(t)$ will asymptotically converge to zero. In addition, the output of the leader can be asymptotically tracked by the output of all followers.*

5.2. Control design for systems with $\rho = 2$

Due to distributed adaptive feedback controller design needs, each follower must acquire the high-order derivative of reference signal $y_0(t)$ with only neighbor information available. Thus, to enable the reference signal with existing high-order derivatives to participate in the controller design, we use $y_{di}(t) = \hat{C}_{i,1}(t)\eta_{i,1}(t)$ as the reference signal by incorporating the auxiliary systems (23)–(26).

In the following, with the parameters B_i, C_i and W_i^* being unknown, the consensus control problem is taken into account.

5.2.1. System model

We can conclude the TCDS of each follower from [Lemma 1](#) as follow:

$$\ddot{y}_i(t) = C_i A_i^2 x_i(t) + C_i A_i W_i^* S_i(x_i(t)) + C_i A_i B_i u_i(t) + C_i W_i^* \dot{S}_i(x_i(t)). \quad (34)$$

Given that $C_i = [c_{i,1}, \dots, c_{i,n}]$, $C_i W_i^* = [v_{i,1}, \dots, v_{i,l}]$, $C_i A_i W_i^* = [p_{i,1}, \dots, p_{i,l}]$, $B_i = [b_{i,1}, \dots, b_{i,n}]^T$ and

$$W_i^* = \begin{bmatrix} w_{i,11}^* & w_{i,12}^* & \dots & w_{i,1l}^* \\ \vdots & \vdots & \ddots & \vdots \\ w_{i,n1}^* & w_{i,n2}^* & \dots & w_{i,nl}^* \end{bmatrix}.$$

Then, we have

$$\begin{aligned} \left(C_i A_i^2 + C_i W_i^* \frac{\partial S_i(x_i)}{\partial x_i} A_i \right) x_i &= [C_i, C_i W_i^*] \left[(A_i^2 x_i)^T, \left(\frac{\partial S_i(x_i)}{\partial x_i} A_i x_i \right)^T \right]^T, \\ C_i W_i^* \frac{\partial S_i(x_i)}{\partial x_i} W_i^* S_i(x_i) &= \sum_{j=1}^l \sum_{k=1}^n \sum_{m=1}^l v_{i,j} w_{i,km} S_{i,m}(x_i) \frac{\partial S_{i,j}(x_i)}{\partial x_{i,k}}, \\ C_i W_i^* \frac{\partial S_i(x_i)}{\partial x_i} B_i u_i &= \sum_{j=1}^l \sum_{k=1}^n v_{i,j} b_{i,k} \frac{\partial S_{i,j}(x_i)}{\partial x_{i,k}} u_i. \end{aligned} \quad (35)$$

Consequently, the system (34) can be parametrized in the form of

$$\ddot{y}_i(t) = \sum_{j=1}^5 \theta_{i,j}^* \omega_{i,j}(t), \quad (36)$$

where $\theta_{i,j}^*, j = 1, \dots, 5$ are the unknown parameters with $\theta_{i,1}^* = [C_i, C_i W_i^*]^T \in \mathbb{R}^{n+l}$, $\theta_{i,2}^* \in \mathbb{R}^{nl^2}$ comprising nl^2 basic elements structured as $v_{i,j} w_{i,km}$, $j, m = 1, \dots, l, k = 1, \dots, n$, $\theta_{i,3}^* = C_i A_i B_i \in \mathbb{R}$, $\theta_{i,4}^* \in \mathbb{R}^{nl}$ comprising nl basic elements structured as $v_{i,j} b_{i,k}$, $j = 1, \dots, l, k = 1, \dots, n$, and $\theta_{i,5}^* = [p_{i,1}, \dots, p_{i,l}] \in \mathbb{R}^l$, and $\omega_{i,j}(t), j = 1, \dots, 5$ are the known regressor vectors such that $\omega_{i,1} = [(A_i^2 x_i)^T, (\frac{\partial S_i(x_i)}{\partial x_i} A_i x_i)^T]^T \in \mathbb{R}^{n+l}$, $\omega_{i,2} \in \mathbb{R}^{nl^2}$ constituted by nl^2 basic elements structured as $S_{i,m} \frac{\partial S_{i,j}(x_i)}{\partial x_{i,k}}, j, m = 1, \dots, l, k = 1, \dots, n$, $\omega_{i,3} = u_i \in \mathbb{R}$, $\omega_{i,4} \in \mathbb{R}^{nl}$ constituted by nl basic elements structured as $\frac{\partial S_{i,j}(x_i)}{\partial x_{i,k}} u_i, j = 1, \dots, l, m = 1, \dots, n$, and $\omega_{i,5} = S_i(x_i) \in \mathbb{R}^l$.

5.2.2. Adaptive controller structure

Considering the form of the parametrized model (36), an adaptive distributed controller can be defined as

$$u_i = \frac{1}{\theta_{i,3} + \theta_{i,4}^T \varphi_i} (\ddot{y}_{di} + \alpha_1 (\dot{y}_{di} - \theta_{i,6}^T \omega_{i,6}) + \alpha_2 (y_{di} - y_i) - \theta_{i,1}^T \omega_{i,1} - \theta_{i,2}^T \omega_{i,2} - \theta_{i,5}^T \omega_{i,5}) \quad (37)$$

where α_1 and α_2 are constant parameters ensuring the polynomial $s^2 + \alpha_1 s + \alpha_2$ is Hurwitz, $\varphi_i = \varphi_i(t) \in \mathbb{R}^{nl}$ is a known vector composed of nl basic elements structured as $\frac{\partial S_{i,j}(x_i)}{\partial x_{i,k}}$, $\theta_{i,j}^*, j = 1, \dots, 5$ are the unknown parameters estimated by $\theta_{i,j}(t), j = 1, \dots, 5$, respectively, $\theta_{i,6} = \theta_{i,6}(t)$ estimates the unknown vector $\theta_{i,6}^* = [C_i, C_i W_i^*]^T$, and $\omega_{i,6} = [(A_i x_i)^T, S^T(x_i)] \in \mathbb{R}^{n+l}$ is a known vector.

5.2.3. Tracking error equation

Inputting (37) into (36) enables the obtaining of

$$\ddot{y}_i = -\tilde{\theta}_i^T \omega_i + \ddot{y}_{di} - \alpha_1 \dot{e}_i - \alpha_2 e_i, \quad (38)$$

where $\theta_i = \theta_i(t) = [\theta_{i,1}^T(t), \theta_{i,2}^T(t), \theta_{i,3}(t), \theta_{i,4}^T(t), \theta_{i,5}^T(t), \theta_{i,6}^T(t)]^T$, $\tilde{\theta}_i = \theta_i - \theta_i^*$ with $\theta_i^* = [\theta_{i,1}^{*T}, \theta_{i,2}^{*T}, \theta_{i,3}^*, \theta_{i,4}^{*T}, \theta_{i,5}^{*T}, \theta_{i,6}^{*T}]^T$, and $\omega_i(t) = [\omega_{i,1}^T(t), \omega_{i,2}^T(t), \omega_{i,3}(t), \omega_{i,4}^T(t), \omega_{i,5}^T(t), \alpha_1 \omega_{i,6}^T(t)]^T$. Thus

$$\ddot{e} + \alpha_1 \dot{e} + \alpha_2 e = \begin{bmatrix} -\tilde{\theta}_1^T \omega_1 \\ \vdots \\ -\tilde{\theta}_N^T \omega_N \end{bmatrix}. \quad (39)$$

For each follower, if the effect exponentially decayed by the initial conditions is ignored, we can rewrite the tracking error [Eq. \(39\)](#) as

$$e_i(t) = -W_m(s) [\tilde{\theta}_i^T \omega_i](t), \quad W_m(s) = \frac{1}{P_m(s)}, \quad (40)$$

where $P_m(s) = s^2 + \alpha_1 s + \alpha_2$ is Hurwitz. Let augmented error be defined as

$$\epsilon_i(t) = e_i(t) - \sigma_i(t), \quad (41)$$

where $\sigma_i(t) = \theta_i^T(t) \zeta_i(t) - W_m(s) [\theta_i^T \omega_i](t)$ with $\zeta_i(t) = W_m(s) [\omega_i](t)$.

5.2.4. Adaptive law

The following gradient-based update law for $\theta_i(t)$ is used:

$$\dot{\theta}_i(t) = \frac{\Gamma \epsilon_i(t) \zeta_i(t)}{m_i^2(t)}, \quad \theta_i(0) = \theta_{i0} \quad (42)$$

where $m_i(t) = (1 + \zeta_i^T(t) \zeta_i(t))^{1/2}$, Γ is a symmetric adaptation gain matrix with $\Gamma > 0$, which serves as the learning rate that governs the speed of parameter adaptation. The initial value for $\theta_i(t)$ is denoted by θ_{i0} . Based on this update law, the following properties can be derived for (42).

Lemma 6. *It is guaranteed by the adaptive law (42) that $\theta_i(t) \in L^\infty$, $\dot{\theta}_i(t) \in L^2 \cap L^\infty$, $(\epsilon_i(t)/m_i(t)) \in L^2 \cap L^\infty$, and $(\epsilon_i(t)/1 + \|\zeta_i\|_t) \in L^2 \cap L^\infty$.*

Proof. Define the Lyapunov function as

$$V(\tilde{\theta}_i) = \frac{1}{2} \tilde{\theta}_i^T(t) \Gamma^{-1} \tilde{\theta}_i(t). \quad (43)$$

When considering (42), the time derivative $\frac{d}{dt} V(\tilde{\theta}_i)$ can be expressed as

$$\dot{V} = \frac{\epsilon_i(t) \tilde{\theta}_i^T(t) \zeta_i(t)}{m_i^2(t)}. \quad (44)$$

Considering (41) and taking (40) into account, it follows that

$$\epsilon_i(t) = -\tilde{\theta}_i^T(t) \zeta_i(t). \quad (45)$$

Then, combining (44) and (45), we obtain

$$\dot{V} = -\frac{\epsilon_i^2(t)}{m_i^2(t)}. \quad (46)$$

This indicates both the non-increasing property of $V(\tilde{\theta}_i(t))$ and the fact that

$$\int_0^\infty \frac{e_i^2(\tau)}{m_i^2(\tau)} d\tau < V(0) < \infty. \quad (47)$$

Consequently, $\theta_i(t) = \tilde{\theta}_i(t) + \theta_i^*$ is bounded. Given the bounded $\tilde{\theta}_i(t)$, $(e_i(t)/m_i(t))$ is bounded as well. As a result, $(e_i(t)/m_i(t)) \in L^2 \cap L^\infty$. Moreover, given $\zeta_i(t)/m_i(t) \in L^\infty$ and referring to (42), it is inferred that $\dot{\theta}_i(t) \in L^2 \cap L^\infty$.

Given the boundedness of $\tilde{\theta}_i(t)$, it follows that $(e_i(t)/1 + \|\zeta_i\|_t) \in L^\infty$. Moreover, noting that $m_i^2(t) \leq (1 + \|\zeta_i\|_t)^2$, we have $\int_0^\infty (e_i^2(\tau)/(1 + \|\zeta_i\|_t)^2) < \infty$, which implies $(e_i(t)/1 + \|\zeta_i\|_t) \in L^2$. This complete the proof. $\square$ $\square$

5.2.5. Performance analysis

Before proceeding with the derivation, we make the following assumption.

Assumption 4. The derivatives of $(\partial S_{i,j}(x_i)/\partial x_{i,k})$, $j = 1, 2, \dots, l$, $k = 1, 2, \dots, n$ with respect to time t are bounded.

After deriving Lemma 6 and making Assumption 4, we can further draw the following conclusions.

Theorem 2. With Assumptions 1 to 4 in effect, by integrating the adaptive disturbance observer (ADO) given by Eqs. (14)–(16) with auxiliary systems (23)–(26) and the adaptive update law (42), the distributed adaptive controller (37) can guarantee that all closed-loop signals of the follower systems (3) with a relative degree of 2 are bounded, and the tracking error $e_i(t) = y_i(t) - y_d(t)$, $i = 1, 2, \dots, N$ between each follower and the leader asymptotically converges to zero.

Proof. Before conducting this derivation, some symbols need to be indicated: c is a generic finite constant, κ and τ are generic $L^2 \cap L^\infty$ functions with κ being arbitrary and $\tau \rightarrow 0$ as $t \rightarrow \infty$. Additionally, $\|q(v)\|_t = \sup_{0 \leq v \leq t} \|q(v)\|$ for $t > 0$, and L^∞ is an extended space of L^∞ , defined as $L^\infty = \{f(t) | \forall s < \infty, f_s(t) \in L^\infty\}$, where $f_s(t) = f(t)$ for $t \leq s$ and $f_s(t) = 0$ for $t > s$. Given that $W_m(s) = c_m(sI - A_m)^{-1}b_m$, by applying the swapping lemma (Sastry & Bodson, 1989, Lemma 3.6.5), it can be deduced that

$$W_m(s)[\tilde{\theta}_i^T \omega_i](t) - \tilde{\theta}_i^T W_m(s)[\omega_i](t) = -c_m(sI - A_m)^{-1}[(sI - A_m)^{-1}b_m[\omega_i^T](t)\tilde{\theta}_i](t). \quad (48)$$

Because $P_m(s)$ is Hurwitz, the stability of $(sI - A_m)^{-1}b_m$ is ensured. Besides, as $\tilde{\theta}_i(t) = \dot{\theta}_i(t) \in L^2 \cap L^\infty$,

$$\|(sI - A_m)^{-1}b_m[\omega_i^T](t)\tilde{\theta}_i\|_t \leq \kappa\|\omega_i\|_t + \kappa. \quad (49)$$

Considering $c_m(sI - A_m)^{-1}$ being strictly proper, it follows that

$$|W_m(s)[\tilde{\theta}_i^T \omega_i](t) - \tilde{\theta}_i^T W_m(s)[\omega_i](t)| \leq \tau\|\omega_i\|_t + \tau. \quad (50)$$

Because $\theta_i(t) \in L^\infty$, and $W_m(s)$ is proper stable, as is $sW_m(s)$, it follows that

$$\|\xi_i\|_t \leq c\|\omega_i\|_t + c, \quad \|\eta_{i,z}\|_t \leq c\|\omega_i\|_t + c. \quad (51)$$

Since $T(x_i) = [\xi_i^T(t), \eta_{i,z}^T(t)]^T$ is a diffeomorphism, it can be concluded that

$$\|x_i\|_t \leq c\|\omega_i\|_t + c, \quad \|\dot{x}_i\|_t \leq c\|\omega_i\|_t + c. \quad (52)$$

Noticing the structure of $\omega_{i,j}$, $j = 1, 2, \dots, 5$ presented below (36), we have

$$\|\omega_i\|_t \leq c\|x_i\|_t + c, \quad \|\dot{\omega}_i\|_t \leq c\|x_i\|_t + c. \quad (53)$$

Combined with (52), it follows that

$$\|\dot{\omega}_i\|_t \leq c\|\omega_i\|_t + c. \quad (54)$$

Because $\zeta_i(t) = W_m(s)[\omega_i](t)$, where $W_m(s)$ is a stable and strictly proper minimum phase, from Lemma 3.6.2 of Sastry and Bodson (1989), we can get

$$\|\dot{\zeta}_i\|_t \leq c\|\zeta_i\|_t + c, \quad (55)$$

which means that the signal ζ_i is regular. Then, using Lemma (Lai et al., 2020, B.3), it is not hard to obtain that

$$|e_i(t)| \leq \tau\|\omega_i\|_t + \tau. \quad (56)$$

Based on (56), we have

$$|e_i(t)| \leq \tau\|\omega_i\|_t + \tau, \quad |\dot{e}_i(t)| \leq \tau\|\dot{\omega}_i\|_t + \tau. \quad (57)$$

Furthermore, combining (52), (53), and (54) gives

$$\left\| \frac{d}{dt}(\tilde{\theta}_i^T \omega_i) \right\|_t \leq c\|\tilde{\theta}_i^T \omega_i\|_t + c. \quad (58)$$

From Lemma 3.6.2 in Sastry and Bodson (1989), we derive the following results:

$$\|\tilde{\theta}_i^T \omega_i\|_t \leq c\|e_i\|_t + c, \quad |\dot{e}_i(t)| \leq \tau\|e_i\|_t + \tau. \quad (59)$$

By contradiction, second inequality in (59) implies $\lim_{t \rightarrow \infty} \|e_i\|_t = 0$. In addition, the property that $\tilde{\theta}_i^T \omega_i(t) \in L^\infty$ can also be derived from (59). Combined with (51) and (52), one has $\xi_i(t), \eta_{i,z}(t), x_i(t) \in L^\infty$, and that $\omega_i(t) \in L^\infty$ by (53). By integrating (37), (52), and the structures of $\omega_{i,1}(t)$, $\omega_{i,2}(t)$ and $\omega_{i,5}(t)$ presented below (36), we can conclude the boundedness of $u_i(t)$. Thus, all closed-loop signals are bounded. $\square$

6. Simulation study

In the following exploration, aiming to demonstrate the effectiveness of the control scheme proposed for the consensus tracking control of a MAS, simulation experiments are carried out. The MAS under study consists of four followers, and some parameters of each agent remain unknown, where all agents have a relative degree of either one or two.

Remark 8. It is worth noting that the numerical example serves as a theoretical verification, whereas the proposed control framework is suited for a broad range of practical applications. In particular, the scheme is applicable to any noncanonical nonlinear system (2) that can be approximated by (3), subject to an appropriate model reformulation.

A representative practical system is the separately excited DC motor (SEDCM), whose full dynamics read

$$T_{i,a}\dot{I}_{i,a}(t) = V_{i,a}(t) - I_{i,a}(t) - \Phi_i(t)\Omega_i(t),$$

$$T_{i,m}\dot{\Omega}_i(t) = \Phi_i(t)I_{i,a}(t) - K_{i,0}\Omega_i(t) - m_{i,L},$$

$$T_{i,f}\dot{\Phi}_i(t) = V_{i,f}(t),$$

$$\Phi_i(t) = \frac{K_{i,1}I_{i,f}(t)}{1 + K_{i,2}I_{i,f}(t)}.$$

Here, $\Omega_i(t)$ denotes the motor angular speed, $I_{i,a}(t)$ the armature current, $\Phi_i(t)$ the magnetic flux, $V_{i,a}(t)$ the armature voltage, and $T_{i,a}, T_{i,m}, T_{i,f}, K_{i,0}, m_{i,L}, K_{i,1}, K_{i,2}$ are constant physical parameters.

Fix the field voltage $V_{i,f} \equiv V_{i,f0}$ to guarantee bounded flux, take armature voltage $u_i(t) = V_{i,a}(t)$ as control input, and define state vector $x_i^o(t) = [\Omega_i(t), I_{i,a}(t), \Phi_i(t)]^T$, output $y_i(t) = \Omega_i(t)$. Rearranging the above dynamics yields

$$\dot{x}_i(t) = f_i^o(x_i^o(t)) + B_i u_i(t), \quad y_i(t) = C_i x_i^o(t),$$

where

$$f_i^o(x_i^o) = \begin{bmatrix} \frac{x_{i,2}^o x_{i,3}^o}{T_{i,m}} - \frac{K_{i,0}}{T_{i,m}} x_{i,1}^o - \frac{m_{i,L}}{T_{i,m}} \\ -\frac{x_{i,2}^o}{T_{i,a}} - \frac{x_{i,1}^o x_{i,3}^o}{T_{i,a}} \\ -\frac{x_{i,3}^o}{(K_{i,1} - K_{i,2} x_{i,3}^o)T_{i,f}} + \frac{1}{T_{i,f}} \end{bmatrix},$$

$$B_i = \begin{bmatrix} 0 \\ 1/T_{i,a} \\ 0 \end{bmatrix}, \quad C_i = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}.$$

This form conforms to Eq. (2) in the manuscript, confirming that SEDCMs fall within the class of systems addressed in this work. $\diamond$

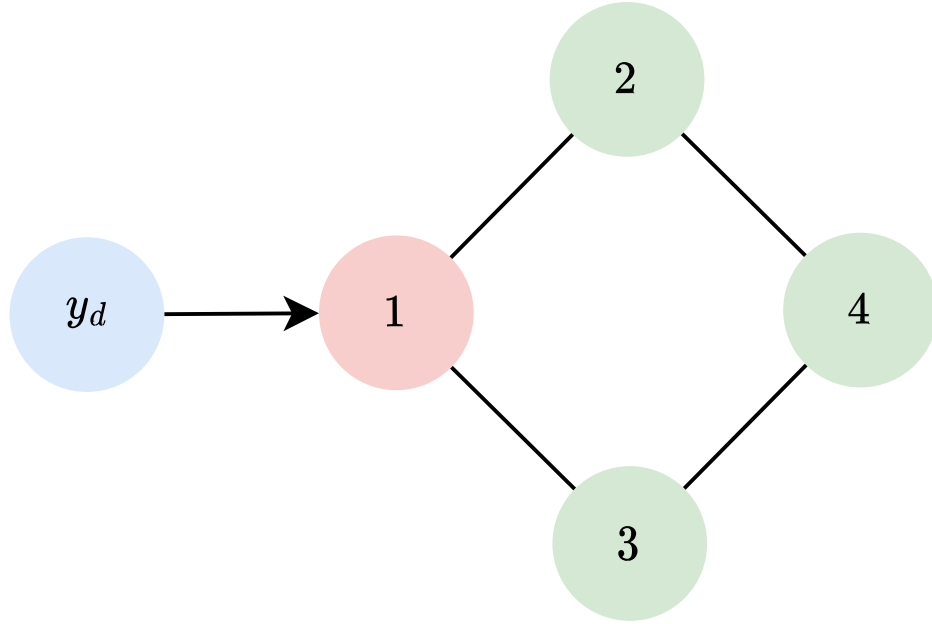
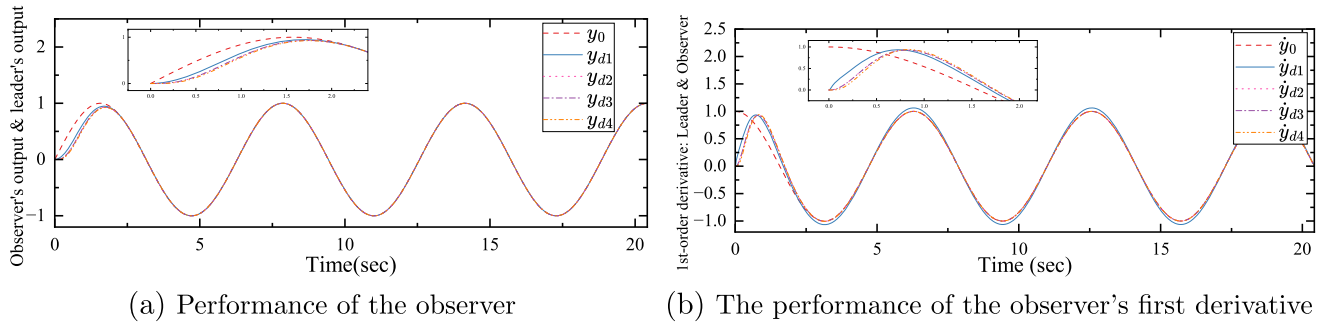


Fig. 1. Communication network.

Fig. 2. When $\rho = 1$, the comparison of observer signal and reference signal.

6.1. Communication topology

The communication map is shown in Fig. 1, and we can obtain the specific values of some matrices, which is

$$\mathcal{A} = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}, \quad \mathcal{P} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

$$\mathcal{D} = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}, \quad \mathcal{L} = \begin{bmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & 0 & -1 \\ -1 & 0 & 2 & -1 \\ 0 & -1 & -1 & 2 \end{bmatrix}.$$

This means that agent 1 receives a reference signal $y_d(t)$ from the leader and is interconnected with two adjacent agents, while the remaining followers are only interconnected with the two adjacent agents.

6.2. System model of agents

For simplicity, in the experimental section, we assume that all four followers are of the same relative degree and adopt the RHONN-based approximation model described in (3). The specific parameters for the four followers are configured as follows: The state vector is $x_i(t) =$

$[x_{i,1}, x_{i,2}]^T \in \mathbb{R}^2$, the input vector is $B_i = [1, 1]^T$. The Hurwitz matrix A_i are set as $A_1 = \text{diag}\{-5, -3\}$, $A_2 = \text{diag}\{-5, -2\}$, $A_3 = \text{diag}\{-5, -1\}$, and $A_4 = \text{diag}\{-10, -1\}$.

The ideal weight matrices W_i^* embedded in the neural approximation term of model (3) are configured for each follower as follows:

$$\begin{aligned} W_1^* &= \begin{bmatrix} 1.68 & 0.94 & 2.2 & 2.55 \\ 2.1 & 1.1 & 2.38 & 3.2 \end{bmatrix}, \\ W_2^* &= \begin{bmatrix} 1.42 & 0.35 & 2.13 & 2.12 \\ 1.72 & 0.85 & 1.8 & 3.1 \end{bmatrix}, \\ W_3^* &= \begin{bmatrix} 1.74 & 1.08 & 2 & 2.14 \\ 3.1 & 0.82 & 1.42 & 2.85 \end{bmatrix}, \\ W_4^* &= \begin{bmatrix} 1.42 & 1.32 & 2.27 & 2.51 \\ 2.1 & 0.78 & 1.54 & 3.6 \end{bmatrix}. \end{aligned} \quad (60)$$

Besides, define a $\text{sig}(\cdot)$ function as $\text{sig}(t) = (1 + e^{-t})^{-1}$. The activation function of agent 1 is selected as

$$S_1(x_1) = [S_{1,1}(x_1), S_{1,2}(x_1), S_{1,3}(x_1), S_{1,4}(x_1)]^T \quad (61)$$

where

$$\begin{aligned} S_{1,1}(x_1) &= s_{1,1}(x_1)s_{1,3}(x_1), \quad S_{1,2}(x_1) = s_{1,1}(x_1)s_{1,2}(x_1), \\ S_{1,3}(x_1) &= s_{1,2}(x_1)s_{1,3}(x_1), \quad S_{1,4}(x_1) = s_{1,3}(x_1)s_{1,4}(x_1), \end{aligned}$$

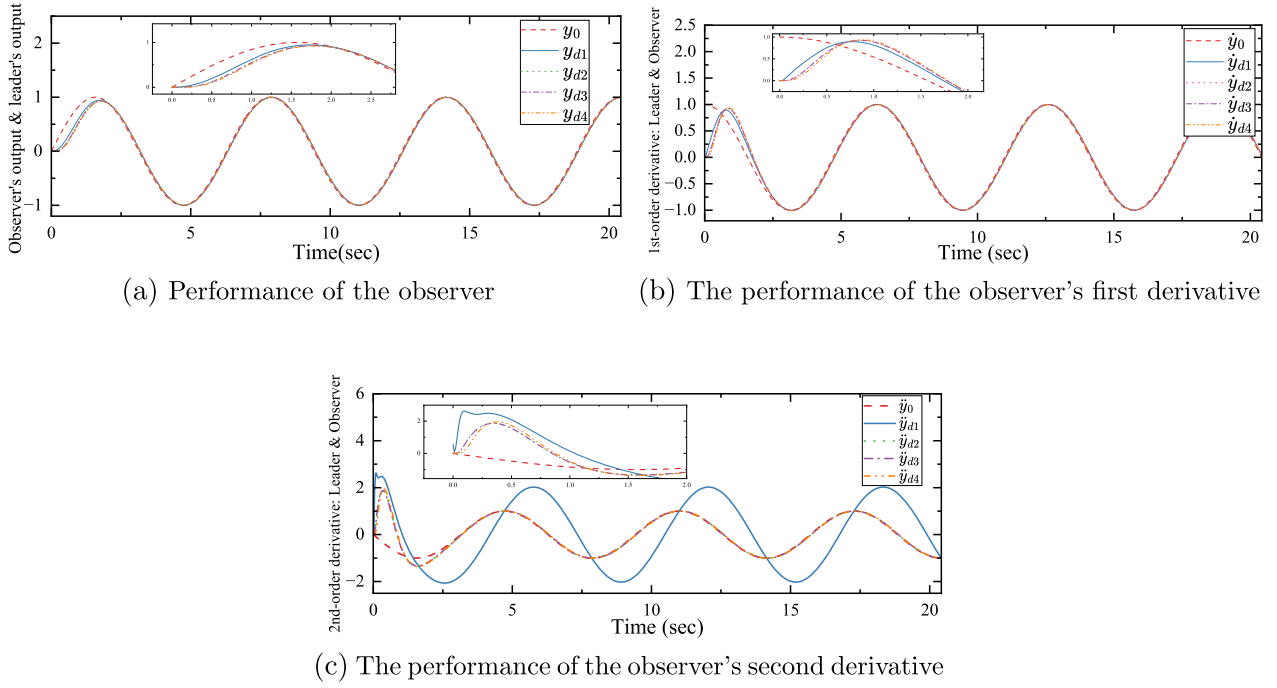


Fig. 3. When $\rho = 2$, the comparison of observer signal and reference signal.

and the specific form of $s_{1,1}(\cdot)$, $s_{1,2}(\cdot)$, $s_{1,3}(\cdot)$, $s_{1,4}(\cdot)$ are

$$\begin{aligned} s_{1,1}(x_1) &= 3\text{sig}(x_{1,1}) - 2, \quad s_{1,2}(x_1) = 2\text{sig}(x_{1,1}) + 1, \\ s_{1,3}(x_1) &= 3\text{sig}(x_{1,1}) - 1, \quad s_{1,4}(x_1) = 2\text{sig}(x_{1,1}) - 0.5 \end{aligned} \quad (62)$$

Similarly, the activation function of agent 2 is

$$\begin{aligned} S_2(x_2) &= [S_{2,1}(x_2), S_{2,2}(x_2), S_{2,3}(x_2), S_{2,4}(x_2)]^T \\ &= [s_{2,1}(x_2)s_{2,2}(x_2), s_{2,1}(x_2)s_{2,3}(x_2), \\ &\quad s_{2,2}(x_2)s_{2,3}(x_2), s_{2,2}(x_2)s_{2,4}(x_2)]^T \end{aligned} \quad (63)$$

with

$$\begin{aligned} s_{2,1}(x_2) &= 2\text{sig}(x_{2,1}) - 1, \quad s_{2,2}(x_2) = 2.5\text{sig}(x_{2,1}) + 1.5, \\ s_{2,3}(x_2) &= 3.5\text{sig}(x_{2,1}) - 1, \quad s_{2,4}(x_2) = 2\text{sig}(x_{2,1}) - 0.5. \end{aligned} \quad (64)$$

The activation function of agent 3 is

$$\begin{aligned} S_3(x_3) &= [S_{3,1}(x_3), S_{3,2}(x_3), S_{3,3}(x_3), S_{3,4}(x_3)]^T \\ &= [s_{3,1}(x_3)s_{3,3}(x_3), s_{3,1}(x_3)s_{3,4}(x_3), \\ &\quad s_{3,2}(x_3)s_{3,3}(x_3), s_{3,2}(x_3)s_{3,4}(x_3)]^T \end{aligned} \quad (65)$$

with

$$\begin{aligned} s_{3,1}(x_3) &= 1\text{sig}(x_{3,1}) - 1, \quad s_{3,2}(x_3) = 1\text{sig}(x_{3,1}) + 1, \\ s_{3,3}(x_3) &= 4\text{sig}(x_{3,1}) - 3, \quad s_{3,4}(x_3) = 0.5\text{sig}(x_{3,1}) + 2. \end{aligned} \quad (66)$$

The activation function of agent 4 is

$$\begin{aligned} S_4(x_4) &= [S_{4,1}(x_4), S_{4,2}(x_4), S_{4,3}(x_4), S_{4,4}(x_4)]^T \\ &= [s_{4,1}(x_4)s_{4,4}(x_4), s_{4,2}(x_4)s_{4,4}(x_4), \\ &\quad s_{4,3}(x_4)s_{4,4}(x_4), s_{4,1}(x_4)s_{4,2}(x_4)]^T, \end{aligned} \quad (67)$$

with

$$\begin{aligned} s_{4,1}(x_4) &= 3\text{sig}(x_{4,1}) - 2, \quad s_{1,2}(x_4) = 3\text{sig}(x_{4,1}) + 1, \\ s_{4,3}(x_4) &= 4\text{sig}(x_{4,1}) - 2, \quad s_{1,4}(x_4) = 4\text{sig}(x_{4,1}) + 1. \end{aligned} \quad (68)$$

In addition, we consider the leader as a linear system, described as

$$\begin{aligned} \dot{x}_0(t) &= S_0 x_0(t) \\ y_0(t) &= C_0 x_0(t) \end{aligned} \quad (69)$$

where $x_0(t) \in \mathbb{R}^2$ and $y_0(t) \in \mathbb{R}$ are the state and output of the leader, respectively, and $S_0 \in \mathbb{R}^{2 \times 2}$ and $C_0 \in \mathbb{R}^{1 \times 2}$ are constant matrices.

In (69), we set

$$S_0 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, C_0 = [1, 0] \quad (70)$$

to generate a sine-wave output signal $y_0(t)$, which serve as the reference signal $y_d(t)$ for the follower.

6.3. Relative degree and feedback linearization problems

6.3.1. Relative degree one

As specified in Section 6.2, we have $C_i B_i \neq 0$. Consequently, the relative degree of all followers is one. In addition, the coordinate transformation matrix is $T(x_i) = C_i x_i$, so the derivative $(\partial T(x_i)/\partial x_i)$ of $T(x_i)$ is nonsingular $\forall x \in \Omega$. Based on the transformation $T(x_i)$, the TCDS is

$$\dot{y}_i = C_i A_i x_i(t) + C_i B_i u_i(t) + C_i W_i^* S(x_i(t)). \quad (71)$$

6.3.2. Relative degree two

Based on the parameter configuration specified in Section 6.2, we can determine that $C_i B_i = 0$, $i = 1, 2, \dots, 4$. For agent 1, based on (61) and (62), we know that $\partial S_{1,1}/\partial x_{1,1} \in (-0.75, 1.5)$, $\partial S_{1,1}/\partial x_{1,2} \in (-1.5, 0.75)$, $\partial S_{1,2}/\partial x_{1,1} \in (-0.02, 1.558)$, $\partial S_{1,2}/\partial x_{1,2} = 0$, $\partial S_{1,3}/\partial x_{1,1} \in (-0.5, 1)$, $\partial S_{1,3}/\partial x_{1,2} \in (0, 2.25)$, $\partial S_{1,4}/\partial x_{1,1} = 0$, $\partial S_{1,4}/\partial x_{1,2} \in (-0.2196, 1.0368)$. Thus, we can further obtain that $C_1 W_1 (\partial S_1 / \partial x_1) B_1 < 1.1809$, which implies that $C_1 A_1 B_1 + C_1 W_1 (\partial S_1 / \partial x_1) B_1 \neq 0$. Therefore, the uniform relative degree of the RHONN approximation system for agent 1 is two on

Ω . Similarly, based on (63)–(68), it can be deduced that the uniform relative degree of agents 2–4 on Ω is also two.

Next, we will choose coordinate transformation

$$T(x_i) = \begin{bmatrix} C_i x_i \\ C_i A_i x_i + C_i W_i^* S(x_i) \end{bmatrix}. \quad (72)$$

For the derivative $(\partial T(x_i)/\partial x_i)$ of $T(x_i)$, we can easily know that $\forall x_i \in \Omega$, it is nonsingular. Thus, $T(x_i)$ is a global diffeomorphism. Through the implementation of the coordinate transformation $T(x_i)$, the TCDS takes the form of

$$\begin{aligned} \ddot{y}_i = & \left(C_i A_i^2 + C_i W_i^* \frac{\partial S_i(x_i)}{\partial x_i} A_i \right) x_i + C_i A_i W_i^* S_i(x_i) \\ & + C_i W_i^* \frac{\partial S_i(x_i)}{\partial x_i} W_i^* S_i(x_i) + \left(C_i A_i B_i + C_i W_i^* \frac{\partial S_i(x_i)}{\partial x_i} B_i \right) u_i. \end{aligned} \quad (73)$$

6.4. Adaptive distributed controller design

6.4.1. Relative degree one

On the basis of the TCDS (71), we design the distributed controller as

$$\begin{aligned} u_i(t) = & \frac{1}{k_{i,3}(t)} \left(-k_{i,1}^T(t) A_i x_i(t) - k_{i,2}^T(t) S_i(x_i(t)) \right. \\ & \left. + \hat{C}_i(t) \eta_i(t) + \hat{C}_i(t) \dot{\eta}_i(t) - a e_i(t) \right) \end{aligned} \quad (74)$$

to achieve consensus tracking, where $e_i(t) = y_i(t) - y_{di}(t)$ and $y_{di}(t) = \hat{C}_i(t) \eta_i(t)$. In the controller, $k_{i,1}(t) \in \mathbb{R}^n$, $k_{i,2}(t) \in \mathbb{R}^l$, and $k_{i,3}(t) \in \mathbb{R}$ serving as respective estimates for C_i^T , $(C_i W_i^*)^T$, and $C_i B_i$.

6.4.2. Relative degree two

On the basis of the TCDS (73), we design the distributed controller as

$$\begin{aligned} u_i = & \frac{1}{\theta_{i,3} + \theta_{i,4}^T \varphi_i} \left(\ddot{y}_{di} + a_1 (\dot{y}_{di} - \theta_{i,6}^T \omega_{i,6}) + a_2 (y_{di} - y_i) \right. \\ & \left. - \theta_{i,1}^T \omega_{i,1} - \theta_{i,2}^T \omega_{i,2} - \theta_{i,5}^T \omega_{i,5} \right) \end{aligned} \quad (75)$$

to achieve consensus control, where $y_{di}(t)$ denotes the observation of the leader's output $y_0(t)$ by the i -th agent. In the controller, $\theta_{i,1} = [\theta_{i,11}, \theta_{i,12}, \dots, \theta_{i,16}]^T \in \mathbb{R}^6$ is the estimate of $[1, -1, w_{i,11}^* - w_{i,21}^*, w_{i,12}^* - w_{i,22}^*, w_{i,13}^* - w_{i,23}^*, w_{i,14}^* - w_{i,24}^*]^T$, $\theta_{i,2} = [\theta_{i,21}, \theta_{i,22}, \dots, \theta_{i,232}]^T \in \mathbb{R}^{32}$ is the estimate of

$$\begin{aligned} & [w_{i,11}^* (w_{i,11}^* - w_{i,21}^*), w_{i,11}^* (w_{i,12}^* - w_{i,22}^*), \\ & w_{i,11}^* (w_{i,13}^* - w_{i,23}^*), w_{i,11}^* (w_{i,14}^* - w_{i,24}^*), \\ & w_{i,21}^* (w_{i,11}^* - w_{i,21}^*), w_{i,21}^* (w_{i,12}^* - w_{i,22}^*), \\ & w_{i,21}^* (w_{i,13}^* - w_{i,23}^*), w_{i,21}^* (w_{i,14}^* - w_{i,24}^*), \\ & \dots \dots \dots \\ & w_{i,24}^* (w_{i,11}^* - w_{i,21}^*), w_{i,24}^* (w_{i,12}^* - w_{i,22}^*), \\ & w_{i,24}^* (w_{i,13}^* - w_{i,23}^*), w_{i,24}^* (w_{i,14}^* - w_{i,24}^*)]^T, \end{aligned} \quad (76)$$

$\theta_{i,3}$ is the estimate of $C_i A_i B_i$, $\theta_{i,4} = [\theta_{i,41}, \theta_{i,42}, \dots, \theta_{i,48}]^T \in \mathbb{R}^8$ is the estimate of $[w_{i,11}^* - w_{i,21}^*, w_{i,11}^* - w_{i,21}^*, w_{i,12}^* - w_{i,22}^*, w_{i,12}^* - w_{i,22}^*, w_{i,13}^* - w_{i,23}^*, w_{i,13}^* - w_{i,23}^*, w_{i,14}^* - w_{i,24}^*, w_{i,14}^* - w_{i,24}^*]^T$, $\theta_{i,5} = [\theta_{i,51}, \theta_{i,52}, \theta_{i,53}, \theta_{i,54}]^T \in \mathbb{R}^4$ is the estimate of $C_i A_i W_i^*$, $\theta_{i,6} = [\theta_{i,61}, \theta_{i,62}, \dots, \theta_{i,66}]^T \in \mathbb{R}^6$ represents the estimate of

$[1, -1, w_{i,11}^* - w_{i,21}^*, \dots, w_{i,14}^* - w_{i,24}^*]^T$, and

$$\begin{aligned} \omega_{i,1} = & [A_{i,11}^2 x_{i,1}, A_{i,22}^2 x_{i,2}, A_{i,11} x_{i,1} \frac{\partial S_{i,1}}{\partial x_{i,1}} + A_{i,22} x_{i,2} \frac{\partial S_{i,1}}{\partial x_{i,2}}, \\ & A_{i,11} x_{i,1} \frac{\partial S_{i,2}}{\partial x_{i,1}} + A_{i,22} x_{i,2} \frac{\partial S_{i,2}}{\partial x_{i,2}}, A_{i,11} x_{i,1} \frac{\partial S_{i,3}}{\partial x_{i,1}} \\ & + A_{i,22} x_{i,2} \frac{\partial S_{i,3}}{\partial x_{i,2}}, A_{i,11} x_{i,1} \frac{\partial S_{i,4}}{\partial x_{i,1}} + A_{i,22} x_{i,2} \frac{\partial S_{i,4}}{\partial x_{i,2}}]^T \\ \omega_{i,2} = & [S_{i,1} \frac{\partial S_{i,1}}{\partial x_{i,1}}, S_{i,1} \frac{\partial S_{i,2}}{\partial x_{i,1}}, S_{i,1} \frac{\partial S_{i,3}}{\partial x_{i,1}}, S_{i,1} \frac{\partial S_{i,4}}{\partial x_{i,1}}, \\ & \dots \dots \dots \\ & S_{i,4} \frac{\partial S_{i,1}}{\partial x_{i,2}}, S_{i,4} \frac{\partial S_{i,2}}{\partial x_{i,2}}, S_{i,4} \frac{\partial S_{i,3}}{\partial x_{i,2}}, S_{i,4} \frac{\partial S_{i,4}}{\partial x_{i,2}}]^T \\ \omega_{i,4} = & [\frac{\partial S_{i,1}}{\partial x_{i,1}}, \frac{\partial S_{i,1}}{\partial x_{i,2}}, \frac{\partial S_{i,2}}{\partial x_{i,1}}, \frac{\partial S_{i,2}}{\partial x_{i,2}}, \frac{\partial S_{i,3}}{\partial x_{i,1}}, \frac{\partial S_{i,3}}{\partial x_{i,2}}, \frac{\partial S_{i,4}}{\partial x_{i,1}}, \frac{\partial S_{i,4}}{\partial x_{i,2}}]^T \cdot u_i \\ \omega_{i,5} = & [S_{i,1}, S_{i,2}, S_{i,3}, S_{i,4}]^T \\ \omega_{i,6} = & [A_{i,11} x_{i,1}, A_{i,22} x_{i,2}, S_{i,1}, S_{i,2}, S_{i,3}, S_{i,4}]^T \\ \varphi_i = & [\frac{\partial S_{i,1}}{\partial x_{i,1}}, \frac{\partial S_{i,1}}{\partial x_{i,2}}, \frac{\partial S_{i,2}}{\partial x_{i,1}}, \frac{\partial S_{i,2}}{\partial x_{i,2}}, \frac{\partial S_{i,3}}{\partial x_{i,1}}, \frac{\partial S_{i,3}}{\partial x_{i,2}}, \frac{\partial S_{i,4}}{\partial x_{i,1}}, \\ & \frac{\partial S_{i,4}}{\partial x_{i,2}}]^T \end{aligned}$$

$\omega_{i,3} = u_i$,

where $A_{i,jk}$, $j, k = 1, 2$ corresponds to the entry in row j and column k of A_i .

6.5. Simulation results

This section validates the distributed controllers (74) and (75) via simulation to prove its effectiveness in the cases where the relative degrees are one and two respectively. As described in (69) and (70) earlier, the followers will track the sinusoidal signal output by the leader.

6.5.1. Case of relative degree one

Based on the Theorem 1, our proposed control scheme, including observer, adaptive parameter estimation, and distributed controller, is capable of achieving output consensus.

In this simulation, we set the initial value of adaptation law (31) corresponding to each follower to $k_i(0) = 0.65k_i^*$, $i = 1, 2, 3, 4$. Additionally, the only coefficient in the controller (74) is set to 12. For simplicity, the parameter settings of the observer are the same as when the relative degree is two, as described in Section 6.5.2.

From the results, the performance of each follower in tracking reference signal $y_0(t)$ is shown in Fig. 4b. Evidently, the proposed control scheme is capable of achieving consensus. Additionally, Fig. 2 illustrates the performance of adaptive distributed observer, including tracking and boundedness. Combining Fig. 4, Fig. 6a, and Fig. 7a, it's clear that all closed-loop signals are bounded.

6.5.2. Case of relative degree two

Based on the Theorem 2 derived from the previous text, the proposed scheme, as is demonstrated, can guarantee closed-loop stability while achieving output consensus for all agents.

In this simulation, we set the initial value of adaptation law (42) corresponding to each follower to $\theta_i(0) = 0.65\theta_i^*$, $i = 1, 2, 3, 4$. It is worth noting that the gains in the observer (14), (15) and (16) are given as $\mu_1 = \mu_2 = 10$ and $\mu_3 = 12$, respectively, satisfying the condition $\mu_1, \mu_2, \mu_3 > \delta_{S_0}^{-1} = 0$.

The performance of each follower in tracking reference signal $y_0(t)$ is shown in Fig. 5b. Evidently, the proposed control scheme is capable of achieving consensus control. Besides, Fig. 3 illustrates the performance of high-order observers, including tracking and boundedness. Combining Fig. 5, Fig. 6b, and Fig. 7b, it's clear that all the closed-loop signals are bounded.

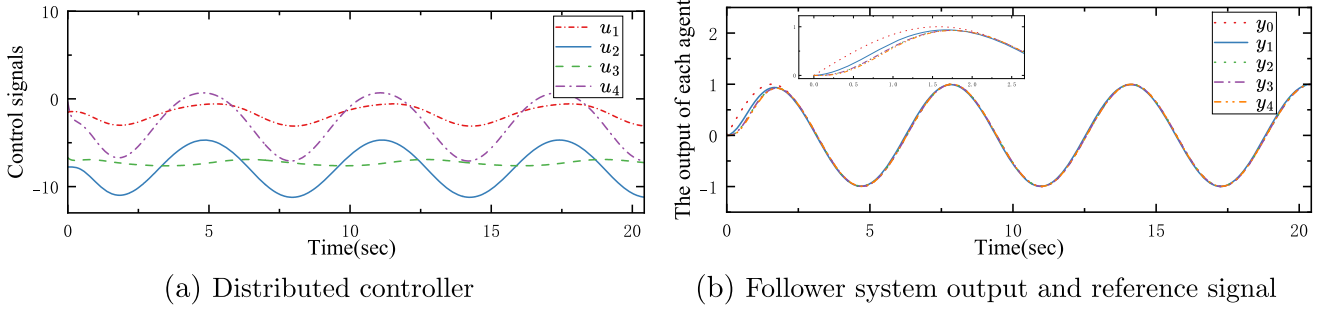
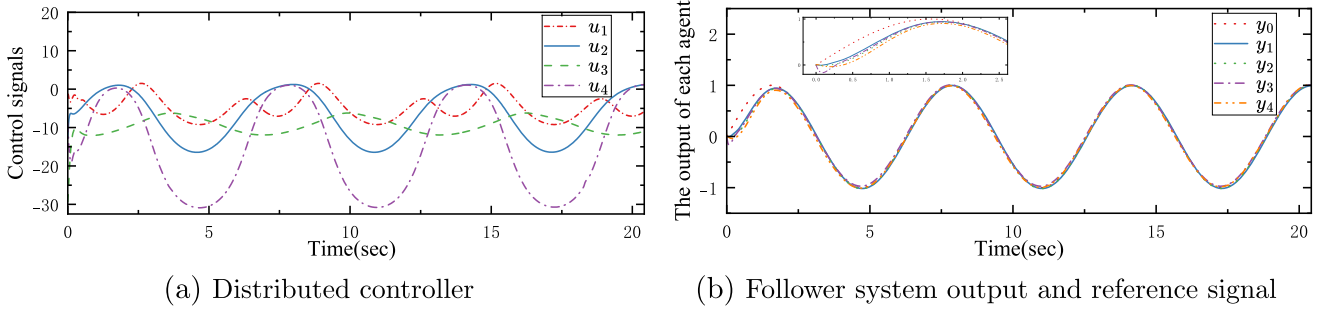
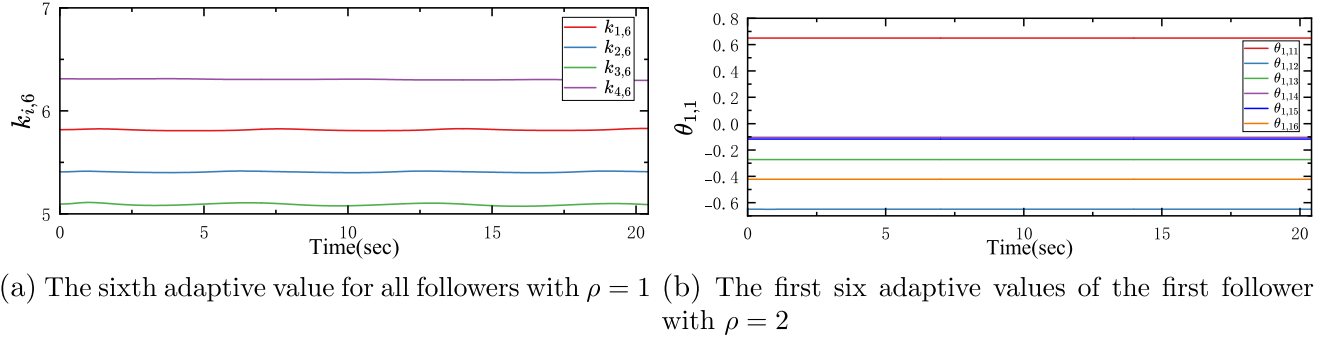
Fig. 4. When $\rho = 1$, the input and output of the follower system.Fig. 5. When $\rho = 2$, the input and output of the follower system.

Fig. 6. Partial estimated variables.

6.6. Comparative experiment with PID control

To further verify the robustness and superiority of the proposed adaptive scheme, a comprehensive comparison with the conventional PID controller is conducted in this section. For brevity, only the tracking performance of Agent 1 is presented. All tests are carried out under the relative-degree-two setting introduced in Section 6.2.

The PID controller is implemented in a distributed fashion. For the i -th follower, the local consensus error is defined as

$$\Delta_i(t) = \sum_{j \in \mathcal{N}_i} a_{ij} (y_i(t) - y_j(t)) + \mu_i (y_i(t) - y_0(t)), \quad (77)$$

where $\mathcal{N}_i$ is the neighbor set of agent i , a_{ij} is the communication weight, and μ_i indicates whether agent i has direct access to the leader signal. The PID control law for the i -th follower is then given by

$$u_i(t) = K_{i,P} \Delta_i(t) + K_{i,I} \int_0^t \Delta_i(\tau) d\tau + K_{i,D} \frac{d\Delta_i(t)}{dt}, \quad (78)$$

where $K_{i,P}$, $K_{i,I}$, and $K_{i,D}$ denote the proportional, integral, and derivative gains for the i -th agent, respectively. Two cases are considered.

Case 1 (Nominal Parameters): The system parameters are configured exactly as specified in Section 6.2. Both the proposed scheme and the PID controller are applied to the nominal system. The PID

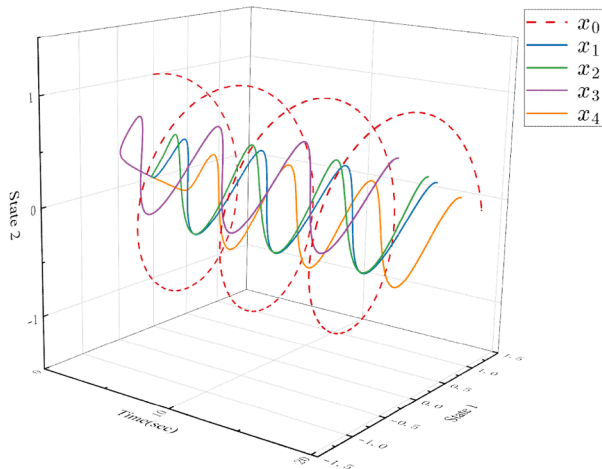
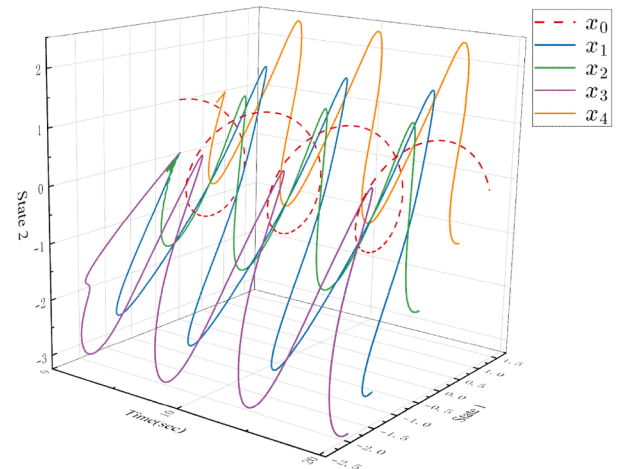
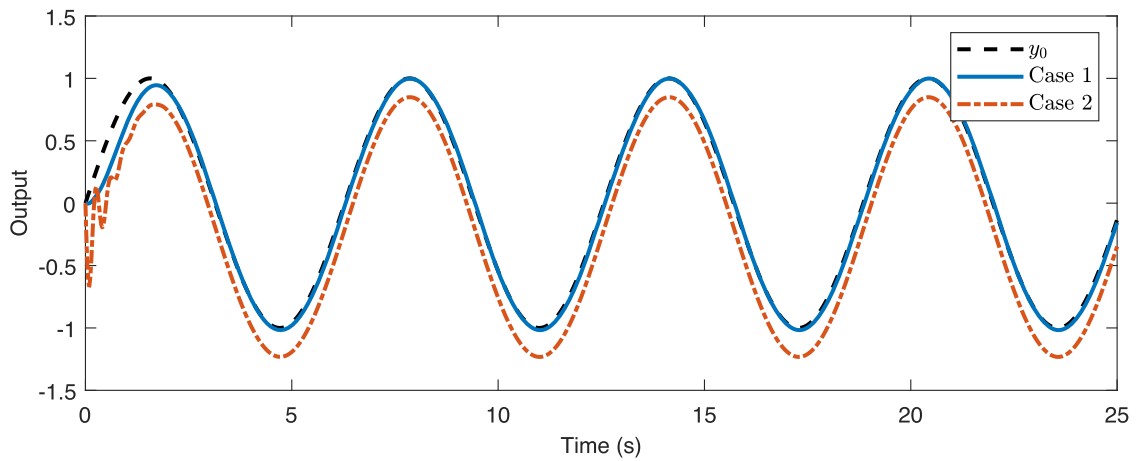
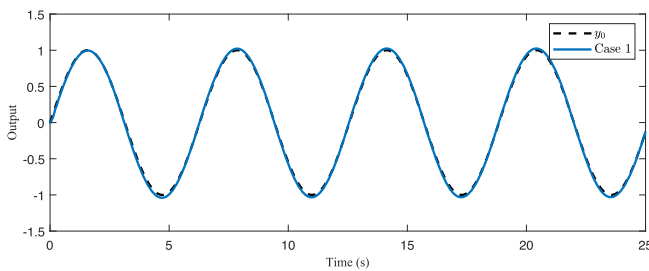
gains are optimized using the particle swarm optimization (PSO) algorithm to achieve the best possible tracking performance under the nominal condition, yielding $K_P = [K_{1,P}, K_{2,P}, K_{3,P}, K_{4,P}] = [-800.0000, -800.0000, -799.3133, -800.0000]$, $K_I = [K_{1,I}, K_{2,I}, K_{3,I}, K_{4,I}] = [-152.2034, -294.8834, -499.1596, -404.9091]$, and $K_D = [K_{1,D}, K_{2,D}, K_{3,D}, K_{4,D}] = [8.1360, -72.0559, -100.0000, -29.5544]$.

Case 2 (Perturbed Parameters): The system parameters are changed without retuning the controllers. Specifically, the weight matrix W_1^* in Eq. (60) is replaced by

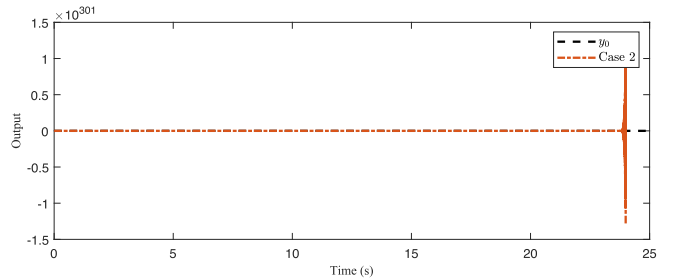
$$W_1^* = \begin{bmatrix} 12.68 & 5.94 & 3.5 & 2.55 \\ 2.1 & 1.1 & 8.38 & 4.2 \end{bmatrix}, \quad (79)$$

while all other system parameters remain unchanged. The controller parameters for both schemes are kept identical to those used in Case 1. This setup simulates a practical scenario where the system dynamics deviate from the nominal model after controller deployment.

Fig. 8 presents the tracking performance of Agent 1 under the proposed scheme in both cases. It can be observed that the output remains stable and continues to track the reference signal despite the parameter perturbation, with only a slight increase in steady tracking error. This demonstrates the inherent robustness of the online adaptive mechanism,

(a) The state of all agents with $\rho = 1$ (b) The state of all agents with $\rho = 2$ **Fig. 7.** The states of agents at different relative degrees.**Fig. 8.** Tracking performance of agent 1 under the proposed scheme.

(a) Performance of PID control in Case 1



(b) Performance of PID control in Case 2

Fig. 9. Tracking performance of agent 1 under the PID controller.

which continuously adjusts the controller parameters to compensate for the altered system dynamics.

Fig. 9a and Fig. 9b depict the tracking performance of the PID controller in Case 1 and Case 2, respectively. Under the nominal parameters, the PID controller with PSO tuned gains achieves fast and accurate tracking, as shown in Fig. 9a. However, when the system parameters are perturbed in Case 2, the PID output diverges and the closed loop system loses stability, as evidenced by Fig. 9b. This is because the PID controller relies on fixed gains tuned for the nominal model and lacks the ability to adapt to changes in system dynamics.

The comparative results highlight a key advantage of the proposed scheme. While both methods perform satisfactorily under nominal conditions, the proposed adaptive controller maintains stability under parameter perturbations that cause the PID controller to fail. Moreover, for nonlinear multi-agent systems, manual tuning or Ziegler–Nichols tuning is infeasible, while PSO-based off-line optimization is time-consuming. In contrast, our scheme features easy parameter tuning without any optimization algorithm. These results confirm the strong robustness of the proposed adaptive scheme against system parameter variations, which is attributed to the

adaptive laws that continuously update the controller parameters on-line.

7. Conclusion

This paper proposes a method to obtain the high-order derivatives from traditional adaptive distributed observers. Based on this method, a control scheme is designed, which achieves the consensus tracking of MASs where the followers have noncanonical nonlinear dynamics. In addition, the stability of the closed-loop system is also assured by our control scheme. Simulation results verify the performance of the high-order adaptive distributed observer and the effectiveness of the proposed control scheme. Comparative experiments further demonstrate that the proposed scheme exhibits stronger robustness against system parameter variations than conventional PID control.

CRediT authorship contribution statement

Guanyu Lai: Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization; **Tao Zhang:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Formal analysis, Data curation; **Xiaohang Su:** Supervision, Project administration, Methodology, Investigation, Formal analysis; **Yuanqing Wu:** Validation, Supervision, Project administration, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Proof of Proposition 1

By assumption, $x_i^o(t) \in \Omega$ for all $t \in [0, T]$, where $\Omega \subset \mathbb{R}^n$ is compact. Define

$$\Omega_\zeta = \{h \in \mathbb{R}^n : \text{dist}(h, \Omega) \leq \zeta\}, \quad (\text{A.1})$$

which is compact and larger than Ω by ζ . The vector $S_i(h)$ is continuous on Ω_ζ . There thus exists $l_s > 0$ such that for all $h_1, h_2 \in \Omega_\zeta$,

$$\|S_i(h_1) - S_i(h_2)\| \leq l_s \|h_1 - h_2\|. \quad (\text{A.2})$$

As shown in Kosmatopoulos et al. (1995), the Stone–Weierstrass theorem implies that if l is sufficiently large, there exist weights $W_i^* \in \mathbb{R}^{n \times l}$ such that

$$\sup_{h \in \Omega_\zeta} \|W_i^* S_i(h) - G_i(h)\| \leq \delta, \quad (\text{A.3})$$

with $\delta > 0$ to be determined.

Define the error $e_i^a(t) = x_i(t) - x_i^o(t)$. The dynamics of the original system can be rewritten as $\dot{x}_i^o = A_i x_i^o + G_i(x_i^o) + B_i u_i$ with $G_i(x_i^o) = f_i^o(x_i^o) - A_i x_i^o$. Subtracting the dynamics of the RHONN model from those of the original system yields

$$\dot{e}_i^a = A_i e_i^a + W_i^* S_i(x_i) - G_i(x_i^o), \quad e_i^a(0) = 0.$$

Adding and subtracting $W_i^* S_i(x_i^o)$ gives

$$\dot{e}_i^a = A_i e_i^a + W_i^* [S_i(x_i) - S_i(x_i^o)] + [W_i^* S_i(x_i^o) - G_i(x_i^o)], \quad e_i^a(0) = 0. \quad (\text{A.4})$$

Assume temporarily that $x_i(t) \in \Omega_\zeta$ for all $t \in [0, T]$. Since A_i is Hurwitz, there exist constants $\kappa \geq 1$ and $\alpha > 0$ such that $\|e^{A_i t}\| \leq \kappa e^{-\alpha t}$. The solution to (A.4) can be expressed as

$$e_i^a(t) = \int_0^t e^{A_i(t-\tau)} (W_i^* [S_i(x_i) - S_i(x_i^o)] + [W_i^* S_i(x_i^o) - G_i(x_i^o)]) d\tau. \quad (\text{A.5})$$

Taking norms and applying (A.2)–(A.3) yields

$$\begin{aligned} \|e_i^a(t)\| &\leq \int_0^t \kappa e^{-\alpha(t-\tau)} (\|W_i^*\| \cdot l_s \|e_i^a(\tau)\| + \delta) d\tau \\ &= \frac{\kappa \delta}{\alpha} (1 - e^{-\alpha t}) + L \int_0^t \kappa e^{-\alpha(t-\tau)} \|e_i^a(\tau)\| d\tau, \end{aligned} \quad (\text{A.6})$$

where $L = \|W_i^*\| l_s$. Let $z(t) = e^{\alpha t} \|e_i^a(t)\|$. Multiplying both sides of (A.6) by $e^{\alpha t}$ gives

$$z(t) \leq \frac{\kappa \delta}{\alpha} (e^{\alpha t} - 1) + \kappa L \int_0^t z(\tau) d\tau.$$

The first term is non-decreasing. The Gronwall–Bellman lemma then ensures that

$$z(t) \leq \frac{\kappa \delta}{\alpha} (e^{\alpha t} - 1) e^{\kappa L t},$$

which leads to

$$\|e_i^a(t)\| \leq \frac{\kappa \delta}{\alpha} (1 - e^{-\alpha t}) e^{\kappa L t} \leq \frac{\kappa \delta}{\alpha} e^{\kappa L T}, \quad \forall t \in [0, T]. \quad (\text{A.7})$$

Choose $\delta = \alpha \zeta e^{-\kappa L T} / \kappa$. Substituting this into (A.7) yields $\|e_i(t)\| \leq \zeta$. By continuity and $e_i^a(0) = 0$, $\|e_i^a(t)\| < \zeta$ holds for all $t \in [0, T]$ and $\zeta > 0$. This completes the proof.

Remark 9. The above result is readily extends to general follower dynamics $\dot{x}_i = F_i(x_i^o, u_i)$ with F_i Lipschitz continuous, by employing the RHONN model $\dot{x}_i = A_i x_i + W_i^T S_i(x_i, u_i)$. Applying the Stone–Weierstrass theorem to $G_i(x_i^o, u) = F_i(x_i^o, u) - A_i x_i^o$ on an augmented compact set in the (x_i^o, u_i) -space, the same approximation guarantee follows. This broadens the applicability of the framework beyond the affine-in-control structure.

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